International Journal of Plasticity 144 (2021) 103059

Contents lists available at ScienceDirect

International Journal of Plasticity

journal homepage: www.elsevier.com/locate/ijplas

A new ANN based crystal plasticity model for FCC materials and its application to non-monotonic strain paths

Olga Ibragimova [a], Abhijit Brahme [a], Waqas Muhammad [a], Julie Lévesque [a,b], Kaan Inal [a,*]

[a] University of Waterloo, 200 University Avenue West, Waterloo, Ontario, N2L 3G1, Canada
[b] Québec Metallurgy Centre, 3095, Rue Westinghouse, Parc Industriel Des Hautes-Forges, Trois-Rivières, Québec, G9A 5E1, Canada

ARTICLE INFO

Keywords:
Machine learning
Artificial neural networks
Crystal plasticity
Non-monotonic loadings
Texture

ABSTRACT

Machine learning (ML) methods are commonly used for pattern recognition in almost any field one could imagine. ML techniques can also offer a substantial improvement in computational time when compared to conventional numerical methods. In this research, a machine learning- and crystal plasticity-based framework is presented to predict stress–strain behaviour and texture evolution for a wide variety of materials within the face-centred cubic family (FCC). Firstly, the process of the framework design is described in detail. The proposed framework was designed to be built of ensemble of artificial neural networks (ANN) and a crystal-plasticity based algorithm. Next, the dataset constituent of crystal plasticity simulations was collected. The dataset consisted of examples of monotonic deformation cases, was prepared for training using mathematical transformations, and finally used to train ANNs used in the framework. Then, the ML framework was demonstrated to predict full stress–strain and texture evolution of different FCC single crystals under uniaxial tension, compression, simple shear, equibiaxial tension, tension–compression–tension, compression–tension–compression, and, finally, for some arbitrary non-monotonic loading cases. The proposed framework predicts the stress–strain response and texture evolution with a high degree of accuracy. The results demonstrated in this research show that the proposed machine learning- and crystal plasticity-based framework exhibits a tremendous computational advantage over conventional crystal plasticity model. Finally, one of the most important contributions of this work is to show the framework's feasibility. The work demonstrates that machine learning methods can help predict complex strain paths without having to train machine learning models on the infinite set of possible non-monotonic loading scenarios.

1. Introduction

Metals are used in a large variety of structural applications and are of high commercial importance. Any metal part, before its production, is modelled and tested to ensure its proper performance. Testing is often carried out by computer-aided engineering, which invokes the usage of complex simulation tools such as finite element (FE) modelling software. Depending on a system, the deformation mechanics can be intricate and can include a large variety of internal state variables in addition to the observable state variables such as geometry, deformation, or temperature (Horstemeyer and Bammann, 2010). While the use of internal state

* Corresponding author.
E-mail address: kinal@uwaterloo.ca (K. Inal).

https://doi.org/10.1016/j.ijplas.2021.103059
Received 22 February 2021; Received in revised form 16 June 2021
Available online 24 June 2021
0749-6419/© 2021 Elsevier Ltd. All rights reserved.

variables can help describe the response mechanisms more clearly (Follansbee and Kocks, 1988; Perzyna, 1986; Roters et al., 2000; Fan and Yang, 2011), it comes at a price of high computational effort and is, as a result, time-consuming.

Crystal Plasticity (CP) models are mesoscopic physics-based models and allow exceptionally accurate estimations of material mechanical response on multiple lengths of scales: from single grains in materials to engineering parts (Roters et al., 2010). In the last thirty years, CP modelling has developed into a very versatile tool that predicts deformation for a wide range of materials such as steel alloys (Zecevic et al., 2016), aluminium alloys (Li et al., 2020), magnesium alloys (Izadbakhsh et al., 2011), titanium alloys (Alankar et al., 2011), and other crystalline materials. CP constitutive equations can be used to incorporate numerous deformation aspects such as dislocation density (Arsenlis and Parks, 2002; Li et al., 2020), grain boundary strengthening (Evers et al., 2004), and dynamic recrystallisation (Popova et al., 2015), and account for material behaviour at elevated temperature conditions (Cyr et al., 2017) and high strain rates (Feather et al., 2019). To account for all of the desired effects, one must include all necessary variables and equations into the constitutive model and then calibrate it for a specific material. Even a basic uniaxial tension CP simulation can be very time consuming (Farooq et al., 2020), as the computational process includes the solution of a number of highly non-linear partial differential equations via numerical methods for hundreds of thousands of integration points. With the complexity involved, CP is a state-of-art method that can help design advanced materials with the exact desired properties for any field of application.

The ability to predict metal deformation accurately and swiftly is crucial, especially when we want to prototype new materials rapidly. Various approaches were investigated to accelerate crystal plasticity calculations, including parallel computing (Inal, 2001), and the wavelet transformation algorithm for accelerated CP simulation of cyclic loading (Joseph et al., 2010; Chakraborty et al., 2011). Kalidindi et al. demonstrated a spectral database method to accelerate crystal plasticity constitutive calculations (Knezevic and Kalidindi, 2017; Gupta et al., 2018; Eghtesad et al., 2018; Kalidindi et al., 2020). The presented database consists of discrete Fourier transforms coefficients that accurately capture the main solutions of CP constitutive equations, thereby avoiding direct computation of solutions. Another work of Kalidindi and co-workers is the Material Knowledge System (MKS) framework — a database approach to help establish linkages within material properties (Kalidindi et al., 2010; Kalidindi, 2015a,b; Brough et al., 2017). MKS employs methods of data science, machine learning, statistics, and other fields to create structure–property–processing relationship for different materials and is used for various applications such as prediction of texture sensitive flow responses (Latypov et al., 2019) and microscale plastic strain rate fields (de Oca Zapiain et al., 2017) in composite materials and other multiscale phenomena in material modelling (Kalidindi, 2012, 2013).

Machine learning (ML) based approaches can be fast once trained and become increasingly accessible. They are designed to capture highly non-linear relations and do not require auxiliary databases for their implementations (Chollet, 2018). Ghaboussi et al. (1991) suggested using artificial neural networks (ANNs) as an alternative to mathematical models of material behaviour and have put the beginning to the applications of machine learning in computational material science. ML methods have been demonstrated to accurately predict various quantities in material science in the past thirty years. Ghaboussi and Sidarta (1998) used ANNs to model the constitutive behaviour of materials. Lefik and Schrefler (2003) used an ANN as an incremental constitutive model implemented in an FE code and demonstrated the predictions of stress paths for a given strain history. A similar prediction approach for flow response was utilised later in other studies (Sabokpa et al., 2012; Haghdadi et al., 2013). Brahme et al. (2009) demonstrated the capability of ANNs to predict cold-rolling fibre texture of steel based on the initial texture data and rolling reduction parameter. Li et al. (2019) presented an ML model that predicted material hardening; this approach was more efficient than the conventional calibration of a yield function. One of the significant studies was conducted by Mozaffar et al. (2019), who demonstrated the prediction of path-dependent material response in terms of stresses and plastic energy for two-dimensional representative volume element (RVE) with a sequence-learning model. Gorji et al. (2020) used several different ANNs to replicate the predictions of anisotropic Yld2000-2d model for various loading conditions. Their worked demonstrated excellent predictive capabilities of feed-forward neural networks and gated-recurrent units network for monotonic and non-monotonic loadings, predicting stress states based on the applied strain. In the recent study, Muhammad et al. (2020) demonstrated an ML framework utilised to predict the evolution of local strain distribution, plastic anisotropy and failure during tensile deformation of 3D-printed aluminium alloy.

Numerous studies involved the utilisation of neural networks and other machine learning algorithms in CP applications. Mangal and Holm (2018b, 2019) used classification ML methods to predict the formation of stress hotspots in polycrystalline materials under uniaxial tension for face-centred cubic (FCC) and hexagonal close-packed (HCP) materials. A dataset of synthetic three-dimensional (3D) microstructures, including microstructural descriptors for each grain, and the stress and strain fields resulting from CP simulations, was also provided by the researchers (Mangal and Holm, 2018a). Ali et al. (2019) proposed an ANN-based framework to predict the flow response and texture evolution of polycrystals subjected to multiaxial and non-proportional loading. They accurately predicted the constitutive response of AA6063-T6 alloy under uniaxial tension followed by simple shear. Other applications of this method included: linear regression and neural networks applied to predict a cyclic stress–strain relation for steels (Miyazawa et al., 2019), sequence learning methods were applied to predict microstructural texture evolution for an FCC copper sample under uniaxial tension (Pandey and Pokharel, 2020), ANNs applied to predict the relation between CP parameters and the microstructural texture for a titanium alloy under uniaxial tension (Acar, 2020), a classification ML algorithm using spatial strain correlation to infer and classify prior deformation history in thin aluminium films (Papanikolaou et al., 2019).

The key to the successful training of an ML model for computational material science applications is to form a meaningful dataset about material behaviour. Most of the mentioned studies highlighted the computational advantage of machine learning models and high accuracy within the bounding box of a training set. The downside of using those models is that their accuracy is typically limited to a range and scope of a training set (Chollet, 2018) as ML typically performs poorly in the extrapolation tasks, although being very good at generalisation. This study will show that it is possible to use machine learning as a tool to achieve accurate

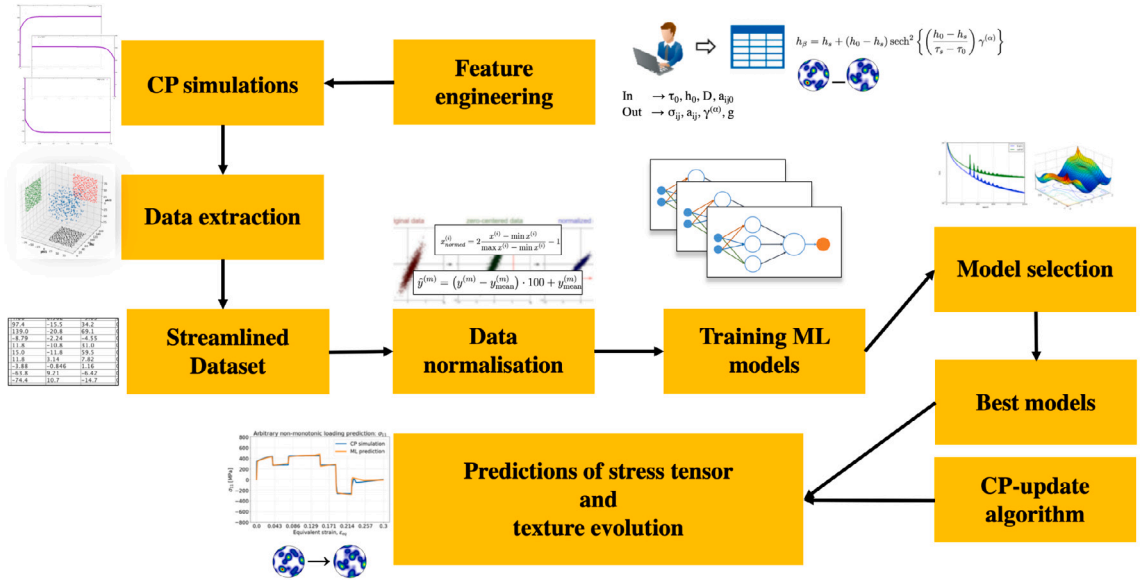


Fig. 1. Schematic representation of the proposed framework.

predictions of complex phenomena such as plastic deformation and texture evolution far beyond the range and scope of a training set.

This paper introduces an approach to couple machine learning and crystal plasticity modelling method. The framework resulting from this approach enables fast and accurate predictions for complex loading for a broad range of face-centred cubic materials. The proposed framework is based on an ensemble of feed-forward neural networks (FFNNs), namely multi-layer perceptrons, trained on monotonic loading scenarios, implemented with a CP update-algorithm. This algorithm allows making predictions far beyond the set of loading conditions included in a training set. It is capable of accurate predictions of stress and texture evolution for a given crystalline material. In addition, the framework predicts evolving value of critical resolved shear stress (CRSS) and accumulated shear as internal variables, which are used to ensure accurate predictions in the cases of non-monotonic loading scenarios.

The organisation of this paper is as follows. The current section presented an introduction to the proposed work. introduces crystal plasticity constitutive model and the development process of the proposed framework. Section 3 presents the predictions obtained with the framework and comparing them with crystal plasticity simulations. Finally, Section 4 summarises the work and concludes the obtained results.

2. Methods & development of machine learning framework

The goal of the present work is to develop and implement a machine learning framework to address the computational drawback associated with crystal plasticity calculations. The proposed framework uses a machine learning- and crystal plasticity-based method that is built upon the data of CP simulations to compute the evolution of stress and texture during deformation under any complex strain path for a wide range of FCC materials. Fig. 1 illustrates the generalised schematics of the proposed framework. The framework involves:

- selection of necessary variables,
- CP simulations for those variables,
- constructing a dataset based on the performed simulations,
- data normalisation to achieve better training results,
- machine learning model selection, training, and evaluation,
- and implementing a CP update algorithm.

The main idea of the implementation of the CP update algorithm, in order, is to (a) identify the evolving parameters in the CP framework, (b) use the trained ML algorithm, calculate the update on the evolving parameters using the obtained predictions, and (c) use the updated parameters as inputs to next the iteration of the ML algorithm.

The next subsections provide details on the CP constitutive model used in this work, dataset generation, the coupling of ML and CP, and artificial neural networks.

2.1. Crystal plasticity constitutive model

Flow behaviour of materials can be modelled using various approaches. Phenomenological plasticity models are most commonly used and are capable of capturing material behaviour under variety of loading conditions (Plunkett et al., 2008; Abedini et al., 2018). However, these models are not designed to predict a texture change within deformed material, in contrast to crystal plasticity models which use crystallographic slip mechanism in their formulations. The growing body of literature demonstrates the power of crystal plasticity models for face-centred cubic (Brahme et al., 2011a; Xiao et al., 2019), body-centred cubic (Inal et al., 2005; Tallman et al., 2017; Zhang et al., 2019), and hexagonal closest packed crystal structure materials (Wang et al., 2013; Paramatmuni and Kanjarla, 2019).

The rate-dependent CP formulation based on the work by Asaro and Needleman (1985) is employed in this work. In this formulation, plastic deformation ensues due to twelve ⟨110⟩ and ⟨111⟩ slip systems. The elastic constitutive equation is expressed as:

$$\overset{\nabla}{\sigma} = \mathcal{L} D - \dot{\sigma}^0 - \sigma \operatorname{tr} D \tag{1}$$

where $\overset{\nabla}{\sigma}$ is the Jaumann rate of Cauchy stress, D is the strain-rate tensor, $\mathcal{L}$ is the elastic stress tensor, and $\dot{\sigma}^0$ is the viscoplastic type stress state.

For each slip system α slip rates are calculated as:

$$\dot{\gamma}^{(\alpha)} = \dot{\gamma}_0 \operatorname{sgn} \tau^{(\alpha)} \left| \frac{\tau^{(\alpha)}}{g^{(\alpha)}} \right|^{1/m} \tag{2}$$

where $\dot{\gamma}_0$ is the reference shear rate, $\tau^{(\alpha)}$ is the resolved shear stress on slip system α, m is the index for strain-rate sensitivity, and $g^{(\alpha)}$ is slip system hardness parameter.

The single crystal work hardening is expressed as:

$$\dot{g}^{(\alpha)} = \sum_{\beta} h_{\alpha\beta} \dot{\gamma}^{(\beta)} \tag{3}$$

where $h_{\alpha\beta}$ are the hardening moduli described as:

$$h_{\alpha\beta} = q_{\alpha\beta} h_{\beta} \quad \text{(no sum on } \beta\text{)} \tag{4}$$

where $q_{\alpha\beta}$ is the term for latent hardening behaviour of a crystal, and h_{β} is the single crystal hardening rate. The term for latent hardening expressed as:

$$q_{\alpha\beta} = \begin{bmatrix} A & qA & qA & qA \\ qA & A & qA & qA \\ qA & qA & A & qA \\ qA & qA & qA & A \end{bmatrix} \tag{5}$$

where A is a 3×3 matrices fully populated with ones, and q is the ratio of latent hardening to self hardening and is set to one.

Finally, single slip hardening used in this work is governed by a hardening law proposed by Chang and Asaro (1981) and is expressed as:

$$h_{\beta} = h_s + (h_0 - h_s) \operatorname{sech}^2 \left\{ \left(\frac{h_0 - h_s}{\tau_s - \tau_0} \right) \gamma^{(\alpha)} \right\} \tag{6}$$

This hardening model depends on the parameters that define material flow behaviour: the initial hardness τ_0, the saturation shear stress τ_s, the initial hardening modulus h_0, and the parameter that controls the slope at the saturation regime h_s.

Following the framework presented by Rossiter et al. (2010), Eqs. (1)–(6) are used to update the stress and the texture. These equations form the core of CP theory and entail a computational drawback associated with CP simulations. The present work aims to enable machine learning to eliminate this computational complexity by achieving reliable predictions comparable to crystal plasticity simulations. The ML framework proposed in this work is used to compute the stress tensor, texture evolution (i.e. rotation matrix), the accumulated shear, and the evolving critical resolved shear stress.

A comprehensive description of this CP model can be observed in the works by Inal et al. (2002) and Rossiter et al. (2010). The presented crystal plasticity constitutive model is used within an in-house crystal plasticity code and was employed to generate the training dataset presented in the Section 2.2.

2.2. Dataset generation

A training dataset has a strong influence on the accuracy of the ML model (Zhang and Ling, 2018). It is vital that the dataset used to construct an accurate machine learning model contains enough information that is representative of a given problem. It could be computationally expensive to collect such a dataset, especially in the case when the model has many input features, as the difficulty of the problem increases with the increase of the number of input features (Jain and Chandrasekaran, 1982; Raudys et al., 1991). Therefore, the input to the model needs to have the minimum amount of features that are necessary to perform accurate predictions,

Table 1
Input features, their ranges, and description of the feature and the physical parameters it controls.

Input variable	Range	Description
τ_0	100.0 … 200.0 MPa	Initial hardness
h_0	70.0 … 500.0 MPa	Initial hardening modulus
D_{ii}, $i = 1 \dots 3$	−1.0 … 1.0	Diagonal components of a strain-rate tensor
D_{12}	−1.0 … 1.0	12-component of a strain-rate tensor
$a_{0_{ij}}$, $i, j = 1 \dots 3$	−1.0 … 1.0	Initial rotation matrix for texture

Table 2
Output features and their description.

Predicted variable	Description
g	Critical resolved shear stress (MPa)
$\sigma_{ij}, i, j = 1 \dots 3$	Stress response (MPa) (6 values, due to symmetrical nature of stress tensor)
$a_{ij}, i, j = 1 \dots 3$	Texture evolution (9 values of rotation matrix)
$\gamma^{(\alpha)}$ (ashear)	Accumulated shear value

and the selection of those features requires the understanding of the field of application. In this case, constructing robust dataset with all necessary information requires understanding of CP and its computational drawbacks.

The process of selecting features out of available data and transformation of those features for a machine learning model is referred to as "feature engineering" (Garla and Brandt, 2012; Seide et al., 2011; Cormack et al., 2007; Severyn and Moschitti, 2013). It is a fundamental process used in ML, as a training dataset has a strong influence on model accuracy. In this work, the process of feature engineering included a selection of a subset of variables from available variable space, which should be sufficient to train an accurate machine learning model f: $f(\text{input}) = \text{output}$. The numerical range for each selected variable was defined and is explained later in this section.

The CP model explained in Section 2.1 was utilised to generate the dataset for training. The model used the hardening law proposed by Chang and Asaro (1981) and is given in Eq. (6). This hardening law offers the most flexibility in capturing the flow behaviour for a wide variety of conditions. The input features used for training the machine learning model include material parameters used in this hardening model viz. initial hardness, initial hardening modulus, as these parameters have a significant influence on the material deformation behaviour. In addition, the input included the strain-rate tensor and initial texture of the crystal. The range on the input parameters is chosen such that it covers a wide variety of material response within the FCC family. Output features of the machine learning framework are the CRSS, stress tensor, texture evolution in the form of rotation matrix, and accumulated shear value. Input and output variables are summarised in Tables 1 and 2 correspondingly. The data was generated for 50 timesteps with $\Delta t = 0.001$ which gives a maximum strain equal to 0.05.

Optimal filling of a feature domain is beneficial for training an optimal model. In this work, no prior knowledge of the feature space is assumed therefore it is advantageous to sample it uniformly. Using the method of grid-sampling would result in a uniform and robust dataset, but it would be very large due to the number of input parameters to the model, and both data-mining and training would be computationally expensive. To overcome this problem and have a robust dataset, Sobol sequences were employed to sample the data. Sobol sequences, also known as LP_τ sequences, have been shown to sample a domain efficiently (Sobol', 1967; Sobol, 1976; Bratley and Fox, 1988; Burhenne et al., 2011). Literature highlights Sobol sequences for their beneficial space-filling properties (Santiago et al., 2012; Bessa et al., 2017). The significant advantage of the sequences is in their low-discrepancy property, i.e. the points in the sequence are almost equidistributed. This fact ensures the domain is filled uniformly but yet quasi-randomly, which is beneficial for training a machine learning model. The algorithm on Sobol sequence generation is explained in Bratley and Fox (1988). The motivation behind utilisation of the Sobol sequence is to generate a sequence (set of numerical points) that is "well-spaced" in the s-dimensional unit cube $I^s = [0,1]^s$, such that:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} f\left(x_i\right) = \int_{I^s} f, \tag{7}$$

where f is some real integrable function over I^s. The Sobol sequence generation algorithm generates data in the range $[0,1]$. Each feature must then be converted to its range. Fig. 2 depicts the visual comparisons of three distinct datasets generated for 343 Euler angles (the number was picked for convenience of the grid sampling approach), and their projections on the coordinate planes $phi1$–Phi, Phi–$phi2$, and $phi1$–$phi2$. The first set was sampled with Sobol sequences, the second – with random sampling from a uniform distribution, and the third one – with grid sampling. One can observe that the first sampling approach provides a more equi-distributed set of points as compared to that generated by the random sampling which may result in clusters of points. These clusters can results in overfitting during training. Finally, grid search does not provide diversity in sampling, in comparison to sampling with Sobol sequences or random sampling, especially in the case of high-dimensional feature space. Therefore, using Sobol sequences in sampling the input data is beneficial for machine learning tasks.

To summarise, the final dataset included $1,451,161$ samples corresponding to unique loading conditions, and the input parameters were sampled using Sobol sequences. Each sample consisted of an input–output set and corresponded to a unique monotonic loading. The total time it took to collect all the data was eight and a half hours for 32 parallel processes, each running on a separate central processing unit core.

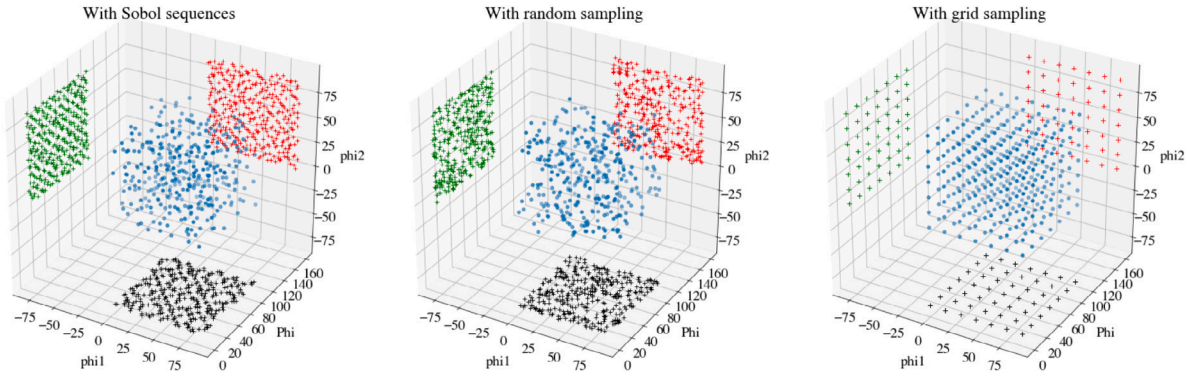


Fig. 2. Visual comparison of sampling with Sobol sequences, random sampling from a uniform distribution, and grid sampling approaches.

Finally, it is very important to efficiently store large datasets. When the available data is small in size, it is often convenient to use standard flat formats such as text (.txt) files, comma-separated values (.csv) files, or excel (.xlsx) files, as such files are easy to open, modify, and access. However, when the data gets more substantial in size, one can hit a roadblock of fitting it into the memory and performing calculations on it. The Hierarchical Data Format (HDF5, where five stands for the version number) technology suite (Folk et al., 2011) is an efficient method of compression and storage of scientific data that is too large to store in memory. This method helps to efficiently organise the large amounts of data, allows compliant and efficient input and output, makes the dataset easily transferable, extensible, and accessible in many frameworks or by the aid of various programming languages including Python (second and third versions), R, FORTRAN, and C++. Within HDF5 data files, the data is stored hierarchically, in forms of multidimensional arrays. The technology suite allows very efficient data compression. A comparison of two files storing the same single-precision floating-point numerical data array of shape $(100000000, 10)$ was performed. The comma-separated data file storing this array used 25.1 GB (GB) of memory, and the HDF5 data file used only 4.0 GB. Data loading time was also significantly improved: it took 18.5 min to access the data from comma-separated value file, and only 590 μs to access data stored in HDF5 format. The latter method of dataset storage was used in the research.

2.3. Coupling machine learning and crystal plasticity

The strength of CP is the ability to predict material response along any complex strain path once it is calibrated for the given material (Ali et al., 2019; Kohar et al., 2019). As mentioned earlier, its main drawback is computational cost, especially when one has to account for complex microstructure with a relatively large number of grains/orientations. Machine learning methods can help eliminate a computational drawback associated with crystal plasticity. The proposed framework couples application of machine learning and crystal plasticity. Training a machine learning model to achieve predictions for any complex strain path, even though possible, is not feasible due to the infinite number of complex loading conditions that will have to be generated, and the model has to be trained on that set. Training an ML model on monotonic loading examples would not result in the model capable of predicting material behaviour under complex non-monotonic strain path because non-monotonic loadings lie in the different distribution of strain paths (Friedman et al., 2001). The challenge is to devise a framework that can predict the response using relatively simple/monotonic strain paths and use those to predict complex strain paths.

In this work, we used the knowledge of CP to develop a framework that is pseudo-recursive in nature. The design of the framework relies on the correct parameterisation of the CP model and the selection of a feature set that encompasses the parameter set presented in Table 1. To perform the correct parameterisation, the identification of CP constitutive model evolving features and their ranges of variation were identified. Then, to predict a result of material deformation under a complex strain path, this complex strain path is to be broken into monotonic parts. The ML algorithm was used to predict output variables for first monotonic part of the strain path and pass them to the CP-update algorithm, receive the updated variables, and repeat the process the needed number of times. The essence of the CP-update algorithm is in keeping the evolving material parameters updated throughout the deformation process like initial hardness τ_0 and hardening modulus h_0. The detailed explanation of prediction process is presented below.

Fig. 3 presents the flowchart that describes the process of prediction for stress and texture evolution for any given strain path. If the strain path of interest is monotonic and the total deformation is less than 0.05 in strain, then the inputs shown in the top left corner of the flowchart (Fig. 3) are sufficient to obtain the predictions for the stress and the evolved texture. The limitation of 0.05 in strain arises from the fact that the implemented ML algorithm was trained on the dataset consisting of monotonic loadings that were 0.05 in strain. It should be mentioned that, for a monotonic strain path of interest less than 0.05 in strain, the predictions of the evolved quantities like $\{g, \gamma^{(\alpha)}\}_{output}$ are inconsequential and only the stress tensor and the texture outputs need to be preserved.

In the case when total strain is larger than 0.05 or in the case when the strain path is non-monotonic, i.e. the strain-rate tensor, D, changes during loading, the following procedure is employed. As the first step, the strain path is split into monotonic parts that are less than 0.05 in strain. For each of the monotonic parts, the prediction procedure is identical to that of the single strain path, i.e. $\{\tau_0, h_0, D, a_{0_{ij}}\}_{input} \rightarrow \{g, \sigma_{ij}, a_{ij}, \gamma^{(\alpha)}\}_{output}$. It should be mentioned that the input of the initial hardness, hardening modulus, and

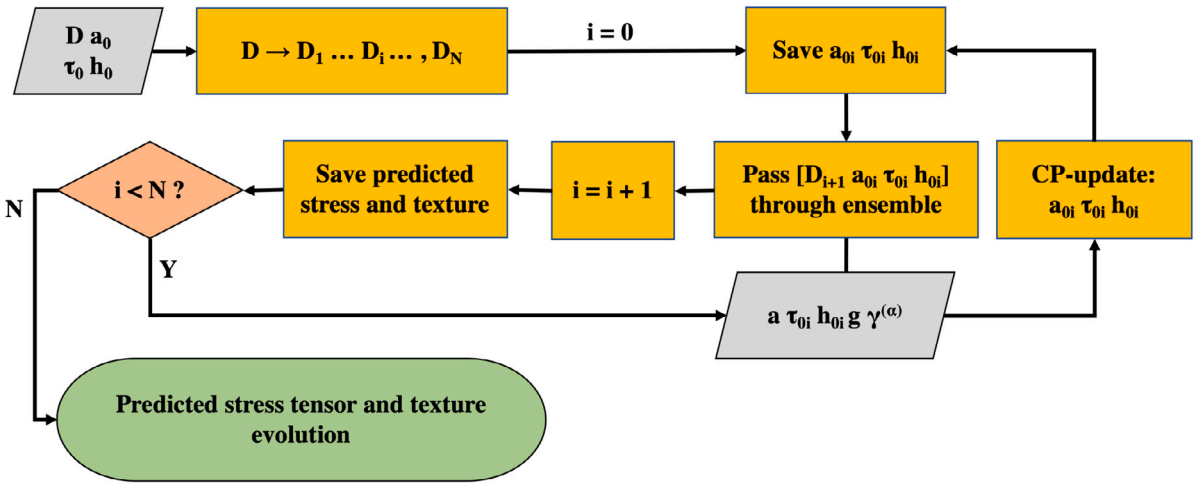


Fig. 3. The flowchart describes the process of obtaining the predictions for a given strain path. Parallelogram units represent the data that is used as input. Rectangle units show processes in the framework. The diamond unit indicates the decision point. Finally, the rounded rectangle unit is the terminal process that outputs final stress–strain and texture evolution predictions.

initial texture for the first monotonic part is identical to the given initial parameters: τ_0, h_0, $a_{0_{ij}}$. Beside the material parameters, the other input is the initial strain-rate D_1. Passing the above input through the ensemble of ANNs produces the following outputs: $\{g_1, \sigma_{1_{ij}}, a_{1_{ij}}, \gamma_1^{(\alpha)}\}_{output}$, updated texture as well as the variables g_1 and $\gamma_1^{(\alpha)}$, which are used to update CP parameters. Note that this procedure is repeated until the desired strain levels are reached for each strain path individually (in parts). As for the CP parameter update, the initial hardness τ_0 is updated with the predicted value of CRSS, g. The predicted values of CRSS exceeding the value of the saturation shear stress, τ_s, are assigned to the value of τ_s. Finally, the hardening modulus is adjusted according to:

$$h_{0_{i+1}} = \begin{cases} h_s + \left(h_{0_i} - h_s\right) \operatorname{sech}^2 \left\{ \left(\frac{h_{0_i} - h_s}{\tau_s - \tau_{0_i}} \right) \gamma_{(i-2)}^{(\alpha)} \right\} & \text{if } g_i < \tau_s \ (i = 1 \dots N \text{ and } \gamma_{(-1)}^{(\alpha)} = 0), \\ h_s & \text{otherwise} \end{cases} \tag{8}$$

where i represents the number of a pass.

The flowchart for the entire procedure is illustrated in Fig. 3, and the used ensemble of ANNs is schematically presented in Fig. 4. Accordingly, an input vector is fed to each network in the ensemble. Each of the networks computes its own feature. The output of the ensemble consists of networks' predictions.

The next subsection presents the details about ANNs used in this framework.

2.4. Artificial neural networks

There are various ANNs currently being adopted. The list of proposed types of ANNs includes, but is not limited to, the multilayer perceptron feed-forward neural networks, convolutional neural networks (CNNs), and recurrent neural networks (RNNs).

Each class of ANNs usually corresponds to a particular kind of problem. For example, FFNNs are commonly used in any field and are also well-known as "universal approximators" (Hornik et al., 1989). FFNNs are typically used with the tabular data and can be applied to a broad spectrum of supervised machine learning tasks (Bishop, 1994). CNN is a class of ANNs commonly used in computer-vision application, object recognition, imagery analysis, and other image-related studies (Abiodun et al., 2018). Note that usage of CNNs is not limited to these tasks. CNNs usually majorly consist of convolutional layers, which are more complex than fully connected layers of FFNNs. RNN is a class of ANNs that is commonly used for the prediction and classification of sequential data of a varied length (such as text or sound) and is noted to have the ability to cope with a broad spectrum of temporal dependencies (Koutnik et al., 2014). Note that the usage of RNNs is not limited to the tasks of sequence learning. There is a number of different sub-types of RNNs, such as Long-Short Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) or Gated Recurrent Unit (GRU) networks (Cho et al., 2014). These networks are more complex than FFNNs in architecture since the connections between the nodes of RNNs form loops, which allow previous outputs to be used as inputs while having hidden states (Keren and Schuller, 2016). In addition, these networks can have variations such as deep RNNs, and Bi-directional RNNs (Graves et al., 2013).

In contrast to CNNs and RNNs, FFNNs are simpler in their architecture (only layers and neurons in hidden layers are subject to variation) and, therefore, easier to be tested in their variety. The data used in the current research was used in the tabular form and related initial material parameters to the corresponding stress–strain response and texture evolution. Hence, the feed-forward

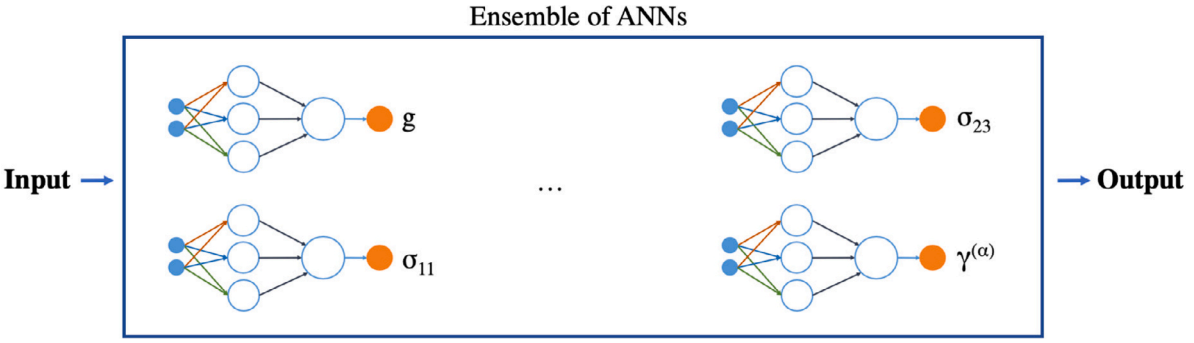


Fig. 4. The figure represents a schematic of ensemble of artificial neural networks. An input vector is fed to each network in the ensemble. Each of the networks computes its own feature. The output of the ensemble consists of networks' predictions.

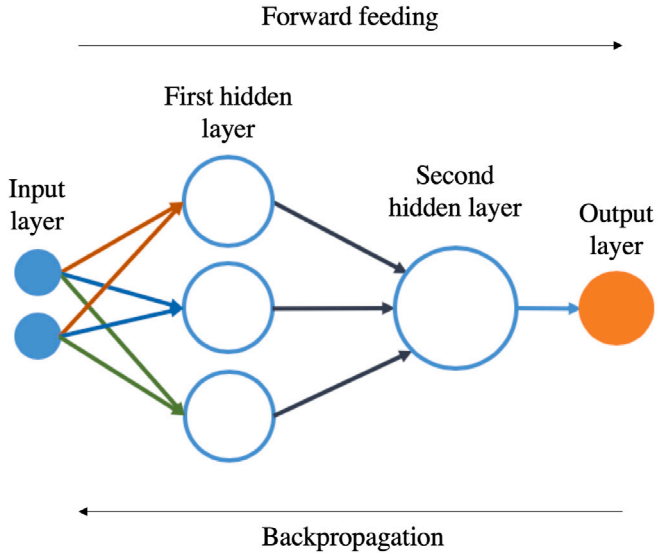


Fig. 5. Schematic representation of a feed-forward artificial neural network with two input unit cells, two hidden layers and one output unit cell. The network has three unit cells in the first hidden layer, and one unit cell in the second hidden layer.

neural network with a backpropagation algorithm was used in this work. From here on, the term "ANN" has been used to refer to feed forward neural network. Fig. 5 demonstrates a schematic representation of such neural network.

The result of training the network is a continuous function, that is formed from a set of interconnected unit cells (Bishop, 2006). The assembly of the unit cells into layers is referred to as network architecture. Each unit cell performs a mathematical transformation of weights and biases and produces an output. The forward-propagation through a feed-forward neural network can be described as:

$$\begin{aligned} a^{[1]} &= g\left(\theta^{[1]}x + b^{[1]}\right) \\ a^{[2]} &= g\left(\theta^{[2]}a^{[1]} + b^{[2]}\right) \\ &\cdots \\ a^{[l-1]} &= g\left(\theta^{[l-1]}a^{[l-2]} + b^{[l-1]}\right) \\ y &= \theta^{[l]}a^{[l-1]} + b^{[l]} \end{aligned} \tag{9}$$

where $\theta^{[l]}$ is a weight matrix in the lth layer, $b^{[l]}$ is a bias vector in the lth layer, $a^{[l]}$ is an output vector in the lth layer, g is an activation function, x is an input vector, and y is an output vector of a neural network. Details about activation functions, normalisation, evaluation, and training of the neural network, used in this work are discussed in further subsections.

Table 3
Features and their normalisation methods.

Variable	Normalisation method
$D_{ij},\ i,j=1\dots3$	No normalisation
τ_0	Normalisation in $-1\dots1$
h_0	Normalisation in $-1\dots1$
g	Normalisation in $-1\dots1$
$\sigma_{ij},\ i,j=1\dots3$	Normalisation in $-1\dots1$
$a_{ij},\ i,j=1\dots3$	Normalisation by range adjustment
$\gamma^{(\alpha)}$	Normalisation in $-1\dots1$

2.4.1. Activation function & normalisation

Activation functions transform a value in a unit cell, typically in a non-linear way (Sharma, 2017). In this study, the hyperbolic tangent activation function was used:

$$\tanh(x)=\frac{(e^{x}-e^{-x})}{(e^{x}+e^{-x})} \tag{10}$$

This activation function is continuous and differentiable and has a range of $[-1,1]$. In the neural network design, this function is often preferred, as its gradients vary in their directions. Furthermore, it is zero centred unlike the sigmoid activation function (Zadeh et al., 2010). In this work the input and output variables were normalised between -1, 1. The hyperbolic tangent activation function performs better with this normalisation. To normalise the ith input or output feature in $[-1,1]$ the following mathematical transformation must be applied to it:

$$x^{(i)}_{normed}=2\frac{x^{(i)}-\min x^{(i)}}{\max x^{(i)}-\min x^{(i)}}-1, \tag{11}$$

where $\min x^{(i)}$ is a minimum and $\max x^{(i)}$ is a maximum value of the ith feature of the input vector or the output vector in the training (i.e. separate from testing and validation parts) dataset.

The other normalisation used in this research was normalisation by range adjustment. We propose this normalisation method to improve the texture evolution ($a_{ij},\ i,j=1\dots3$) learnability. Since all components in the orientation matrix fall within the range of $[-1,1]$, normalisation is not required. However, the statistical range (i.e. the difference between the largest and smallest values) of each output curve a_{ij} is very small. Maximal statistical ranges of each curve do not exceed 0.03. Therefore, to improve the learnability of texture evolution, it is beneficial to adjust their range by its enlargement. To enlarge the statistical range for each curve by the factor of 100, the following mathematical expression was applied:

$$\hat{y}^{(m)}=\left(y^{(m)}-y^{(m)}_{\text{mean}}\right)\cdot100+y^{(m)}_{\text{mean}}, \tag{12}$$

and the backward transformation is described by:

$$y^{(m)}=\left(\hat{y}^{(m)}-\hat{y}^{(m)}_{\text{mean}}\right)/100+\hat{y}^{(m)}_{\text{mean}}, \tag{13}$$

where m is a number of samples in a dataset. Note, that the mean value for the original and the transformed curves will be the same (i.e. $\hat{y}^{(m)}=y^{(m)}$ for each m). Further normalisation is not required, as the last layer's normalisation function is a linear (pass-through) activation function and can predict the values that lie outside the $[-1,1]$ range.

Normalisation for input strain rate tensor is not required, as all its values fall within the $[-1,1]$ range. Table 3 summarises the normalisation methods for each variable.

2.4.2. Evaluation of artificial neural networks

The cost function, $J(\theta,b)$, and loss function, $\mathcal{L}\left(y^{(i)}_{pred},y^{(i)}_{true}\right)$, measure the "goodness" of the trained network. The cost function measures the performance of the model on the dataset. The value of the function is minimised by an optimisation algorithm during training by updating the weights' and biases' values in the "right" direction (Chollet, 2018). The loss function measures how well the model performs on a single training example. Cost function depends on the loss function as follows:

$$J(\theta,b)=\frac{1}{m}\sum_{i=1}^{m}\mathcal{L}\left(y^{(i)}_{pred},y^{(i)}_{true}\right), \tag{14}$$

where m is a number of samples in an evaluated dataset.

Mean squared error (MSE) is a common loss function for regression problems and is used as both loss function and evaluation metrics in this research. The formula for MSE is the following:

$$\mathcal{L}_{\text{MSE}}\left(y^{(i)}_{pred},y^{(i)}_{true}\right)=\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)}_{true}-y^{(i)}_{pred}\right)^{2}, \tag{15}$$

where n is a number of elements in an output vector.

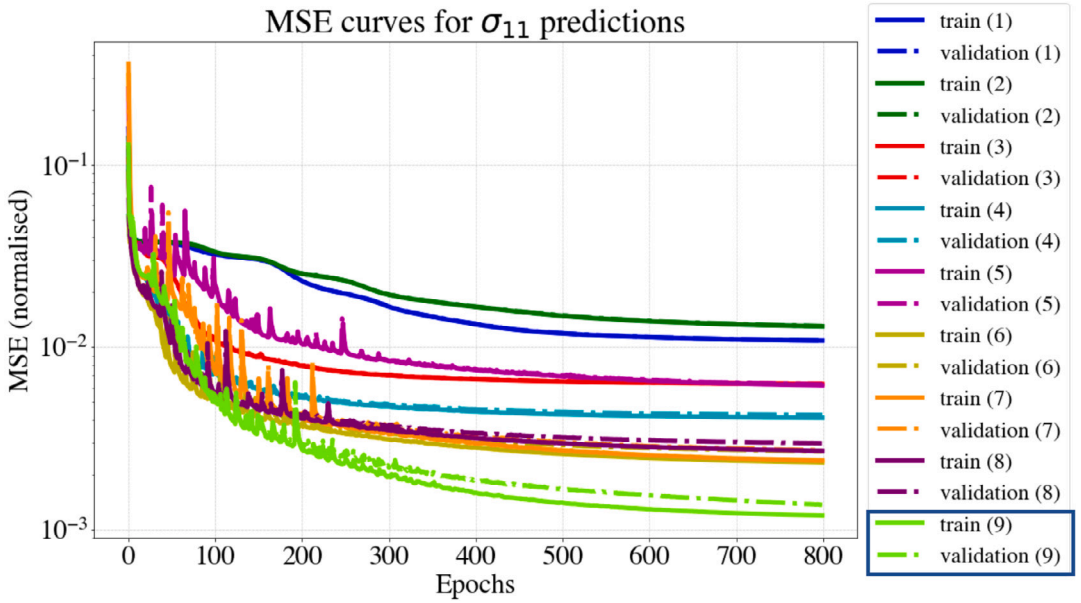


Fig. 6. Neural networks learning rates showing the evolution of losses for training and validation datasets. Each network is numerated, and each number corresponds to unique network architecture. For each ANN, the architecture and validation error are summarised in Table 4.

2.4.3. Optimisation algorithm

Adaptive Moment Estimation (Adam) (Kingma and Ba, 2014) is one of the recently developed gradient descent-based backpropagation optimisation algorithms and is commonly used. This algorithm was used for training ANNs. The algorithm computes adaptive learning rates for each parameter. It stores an exponentially decaying average of past squared gradients v_t like algorithms such as Nesterov's accelerated gradient method (Nesterov, 1983), MaxProp (Burgess et al., 2006), AdaDelta (Zeiler, 2012), and others. It differs in the way it updates an exponentially decaying average of past gradients:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + \left(1-\beta_1\right) g_t, \\ v_t &= \beta_2 v_{t-1} + \left(1-\beta_2\right) g_t^2, \end{aligned} \tag{16}$$

where m_t and v_t are approximations of the first moment (the mean) and the second moment (the non-centred variance) of the gradients. Bias-corrected first and second moment estimates are computed to counter moments being biased towards zero:

$$\begin{aligned} \hat{m}_t &= \frac{m_t}{1-\beta_1^t}, \\ \hat{v}_t &= \frac{v_t}{1-\beta_2^t}. \end{aligned} \tag{17}$$

Eventually, parameters are updated according to:

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{\hat{v}_t}+\epsilon}\hat{m}_t, \tag{18}$$

where β_1, β_2, α, and ϵ are the algorithm hyperparameters and are subject to tuning.

2.4.4. Artificial neural networks architectures

In order to select ANNs for the proposed framework, several networks with different architectures were compared in their performance. The effect of the total number of layers and the number of neurons per layer on the accuracy of the network was studied. Fig. 6 demonstrates the evolution of the mean squared errors for training and validation dataset during training for different architectures of the model for prediction of a σ_{11}-stress component. The network number, in the legend of the figure, corresponds to a different architecture which are presented in Table 4 along with the corresponding validation errors. It can be noted, that the ANNs (5) and (6) have the same number of hidden layers and neurons within hidden layers, but the sequence of hidden layers is different. It can be observed that the network (6) which has a descending number of unit cells from input to output is more accurate than the network in which unit cells in hidden layers ascend from input to output. It is observed that the training and validation error decreases proportionally to the depth on the network. The network (9) is a deep neural network and has 15 hidden layers, and it is the most accurate out of all other networks. An ANN with this architecture was chosen for prediction of σ_{11} stress tensor component based on these observations.

Table 4
The architectures used in the architecture search for training an artificial neural network to predict a σ_{11} component of a stress response.

Architecture number	Architecture	Validation MSE
(1)	[512]	$1.09E{-}02$
(2)	[1024]	$1.31E{-}02$
(3)	[1024 – 512]	$6.30E{-}03$
(4)	[1024 – 512 – 512 – 26]	$4.25E{-}03$
(5)	[128 – 256 – 512 – 1024 – 1024 – 2048]	$6.25E{-}03$
(6)	[2048 – 1024 – 1024 – 512 – 256 – 128]	$2.69E{-}03$
(7)	[4096 – 2048 – 1024 – 1024 – 1024 – 512 – 512 – 512]	$2.72E{-}03$
(8)	[1024 – 1024 – 1024 – 512 – 512 – 512 – 256]	$2.97E{-}03$
(9)	[2048 – 1024 – 1024 – 512 – 512 – 512 – 512 – 256 – 256 – 256 – 256 – 128 – 128 – 128 – 128]	$1.37E{-}03$

Table 5
Final architectures for the networks implemented in the framework. Architecture column displays the number of unit cells in each hidden layer. Test RMSE columns presented the error on the test set prediction.

Predicted variable	Architecture	Test RMSE
$\sigma_{ij},\ i,j=1\dots3$	[2048 – 1024 – 1024– 512 – 512 – 512– 512 – 256 – 256– 256 – 256 – 128– 128 – 128 – 128]	σ_{11}: $8.78E{-}02$ [MPa] σ_{22}: $8.34E{-}02$ [MPa] σ_{33}: $2.80E{-}02$ [MPa] σ_{12}: $4.34E{-}02$ [MPa] σ_{13}: $5.13E{-}02$ [MPa] σ_{23}: $1.02E{-}01$ [MPa]
$a_{ij},\ i,j=1\dots3$ g $\gamma^{(\alpha)}$	[1024 – 1024 – 1024– 512 – 512 – 512– 256 – 256 – 256]	a_{11}: $2.87E{-}02$ a_{12}: $3.06E{-}02$ a_{13}: $2.21E{-}02$ a_{21}: $1.98E{-}02$ a_{22}: $3.14E{-}02$ a_{23}: $2.72E{-}02$ a_{31}: $2.62E{-}02$ a_{32}: $1.87E{-}02$ a_{33}: $3.09E{-}02$ g : $1.04E{-}02$ [MPa] $\gamma^{(\alpha)}$: $1.36E{-}02$

For brevity, process of model selection is shown only for σ_{11} stress tensor component. Similar procedure was performed for all other output variables, and the most accurate models were incorporated into the framework. Final architectures for each model are reported in Table 5.

To train neural networks, the available dataset was split into three parts: training set, validation set, and test set. Training set consisted of 80% of all data, the validation and test sets consisted of 10% of all data, each. Table 5 reports the test set root mean squared error for each ANN, and each error is converted back to the scale of each variable. The formula for RMSE is demonstrated by Eq. (19), where N is the number of observations in a test set, y_i^{CP} is the prediction by crystal plasticity simulation, and y_i^{ANN} is the prediction by an artificial neural network, and i is the number of the sample in the test set.

$$RMSE = \sqrt{\sum_{i=1}^{N} \frac{\left(y_i^{CP} - y_i^{ANN}\right)^2}{N}} \tag{19}$$

For the stress–strain, texture, CRSS, and accumulated shear predictions, the test set prediction errors lies far within the 1% error gap, thus confirming the accuracy of each ANN.

Each artificial neural network was trained for a maximum of 800 training iterations. Initial learning rate, α, was 10^{-3}, decreasing by a factor of 0.95 in the case of no training progress being made over five consecutive training epochs (i.e. number of passes of the training set). The other optimiser hyperparameters were set to their standard values: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$. For each training, early stopping was used to stop training if the change in learning metrics did not exceed 10^{-8} over ten consecutive training iterations. The batch size (i.e. the number of training examples used in one iteration) of 2^{16} was used for training the networks. An approximate time to train each network was 150 minutes. After test errors validated the accuracy of ANNs, the test and validation set was also shuffled into training data, and the models were further trained for another 50 epoch. Finally, the obtained ANNs were used in the framework to predict the response and the computational inefficiency is compared with the CP model.

3. Results & discussion

As discussed earlier, this framework is capable of predicting full stress–strain behaviour for any strain path and for any crystal orientation. This section presents predictions of the framework and verifies the framework performance against CP simulations.

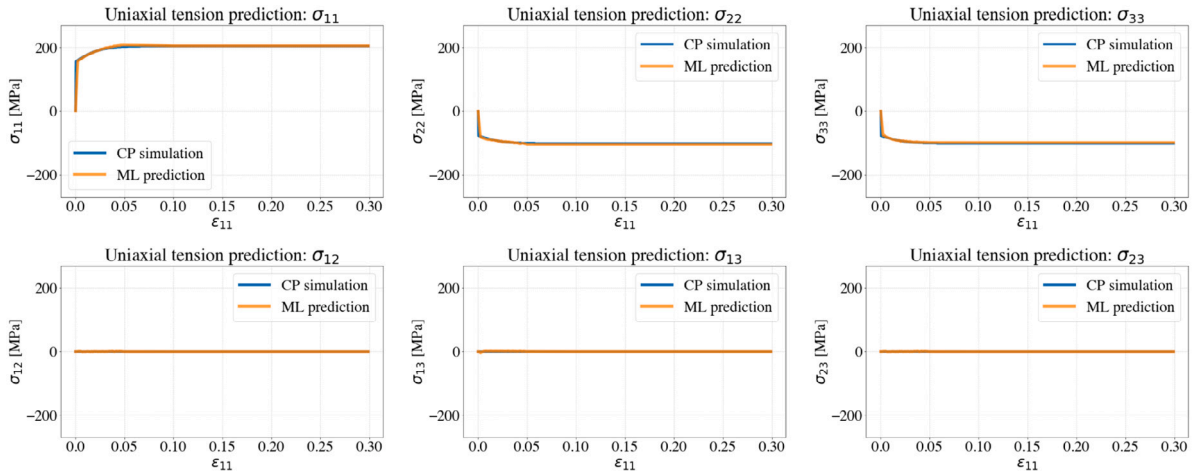


Fig. 7. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) single crystal stress–strain curves under uniaxial tension.

Crystal plasticity simulations were computed for different strain paths, both monotonic and non-monotonic, for different single crystal orientations as well as polycrystal ensembles, and were used to verify the predictions of the framework. In this section, the errors are quantified with the RMSE metrics, as it provides a comprehensive single number performance evaluation given in the same units as the measured variables. The RMSE was calculated across all points on each flow curve and then was averaged across all stress components. The RMSE for texture evolution was calculated using RMSE for each predicted timestep for each Euler angle, and then was averaged across all Euler angles. All the predictions are shown graphically in figures. The subsequent subsections present the results for:

3.1 monotonic strain path predictions
3.2 non-monotonic strain path framework predictions
3.3 polycrystal deformation predictions for monotonic and non-monotonic strain paths

Finally, a runtime comparison between CP and the ANN was performed and is discussed in Section 3.4.

3.1. Predictions for monotonic strain paths

Once trained the ANN framework can be used to predict the flow behaviour for any arbitrary texture for any strain path. This subsection will demonstrate the ability of the framework to predict the texture evolution and as well stress evolution during monotonic loading. In addition, this will also demonstrate the ability of the framework to predict the evolved texture and stress state outside the bounds initial training dataset. To illustrate the predictive capability different single texture components are chosen for each of the monotonic strain paths. The following single crystal orientations are used: Cube (0°, 0°, 0°) under uniaxial tension and simple shear, Goss (0°, 45°, 0°) under compression, and Copper (90°, 35°, 45°) under equibiaxial tension. For each strain path the total strain prescribed was equal to 0.3, except for the case of simple shear where the prescribed strain was equal to 0.4 for demonstration of texture prediction framework's capabilities. Note that the training data was limited to include a maximum of 0.05 strain.

Figs. 7 and 9 show the uniaxial tension and uniaxial compression stress–strain predictions for a Cube-oriented and for the Goss-oriented crystal for crystal plasticity and the ANN framework respectively. The framework predictions show excellent agreement with CP simulation results. The total RMSE for the uniaxial predictions is 1.64 MPa, while that for the compression predictions is 2.02 MPa which confirms the accuracy of the framework. The error is also indicated in Table 8.

Figs. 8 and 10 show the texture evolution for the cases of uniaxial tension and uniaxial compression respectively. During uniaxial tension and compression there is no significant texture evolution (Pandey et al., 2013; Ali et al., 2017) and, in both cases, this is accurately captured by the framework. The texture evolution prediction error is 10^{-4} degrees for both tension and compression; the error is reported in Table 8.

In order to obtain better understanding of framework capability to predict texture evolution, shear test was performed. Shear deformation adds more complexity in comparison to uniaxial tension and compression, as it significantly changes the texture throughout the deformation (Muhammad et al., 2017). In total, 0.4 shear strain was prescribed to Cube single crystal. The evolution of texture is seen as a clockwise rotation of texture as shown in Fig. 12. The framework prediction is in excellent agreement with CP simulation. The prediction error for texture was evaluated at 0.38°, thus confirming the accuracy of the framework.

Stress–strain response prediction for simple shear was also performed and is displayed on Fig. 11. The stress–strain predictions are also in good agreement with crystal plasticity simulation. The overall error of predictions is 1.94 MPa and is reported in Table 8.

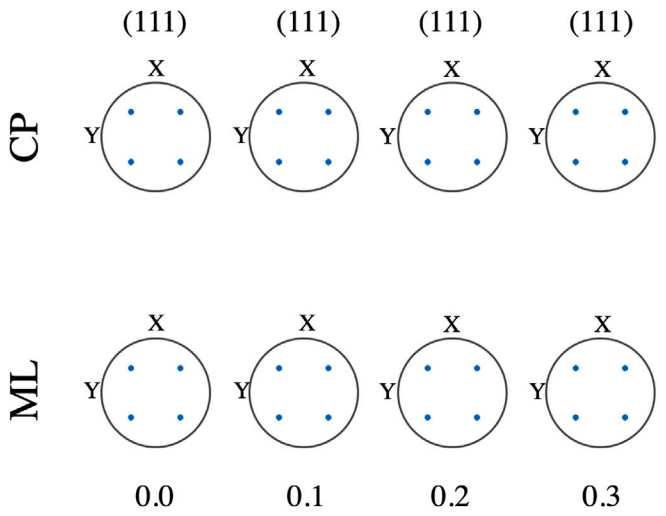


Fig. 8. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) Cube crystal ⟨111⟩ pole figures displaying texture evolution under uniaxial tension. The result comparison is indicated for 0.0 (initial texture), 0.1, 0.2, and 0.3 values of ε_{11} engineering strain.

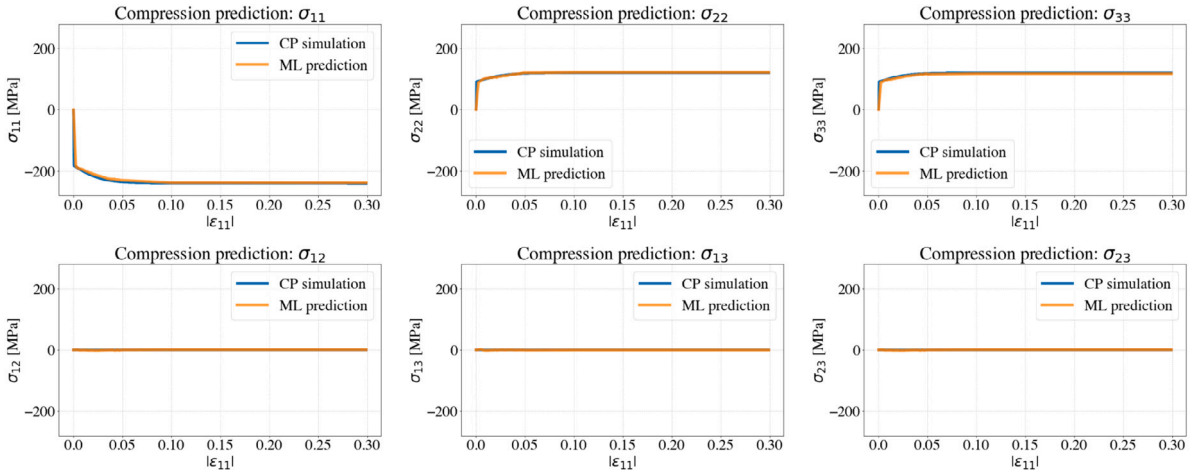


Fig. 9. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) single crystal stress–strain curves under compression.

It is also noted, that simple shear stress–strain behaviour is more complex than in the case of tension and compression. The applied shear is in 12 direction. The framework captures sudden changes of stress in the end of the deformation for σ_{11} and σ_{22} components, as highlighted on the plot. Further, the framework was also able to predict the decrease in stress for σ_{12}. Most importantly, the shear stress σ_{12} was captured exceptionally well. Thus, the framework accuracy was confirmed as in the cases of uniaxial tension and compression.

The other strain path that has prominent effects on texture evolution is equibiaxial tension (Banovic et al., 2008), and hence is also used to validate the framework performance. As in other cases, the predictions for stress–strain behaviour and texture evolution were performed. Fig. 13 shows the crystal plasticity and the framework equibiaxial tension stress–strain predictions for a Copper crystal up to 0.3 for ε_{11} and ε_{22}. The overall stress–strain prediction error was evaluated at 2.47 MPa, and is reported in Table 8. The framework was capable to capture the sudden stress changes in the beginning of deformation for σ_{22} and σ_{13} components of stress. Predictions for σ_{12} and σ_{23} are at zero, as the CP simulations for these components. The framework also exhibits remarkable accuracy in prediction of subtle decrease in stress for σ_{11} component and the increase for σ_{33} component, thus confirming the accuracy of predictions.

The prediction of texture evolution for the case of equibiaxial tension is exhibited on Fig. 14. The overall error for predictions was evaluated at 0.63° and is reported in Table 8. It can be seen that the texture undergoes some rotation, and is successfully captured by the proposed framework, thus, confirming its accuracy.

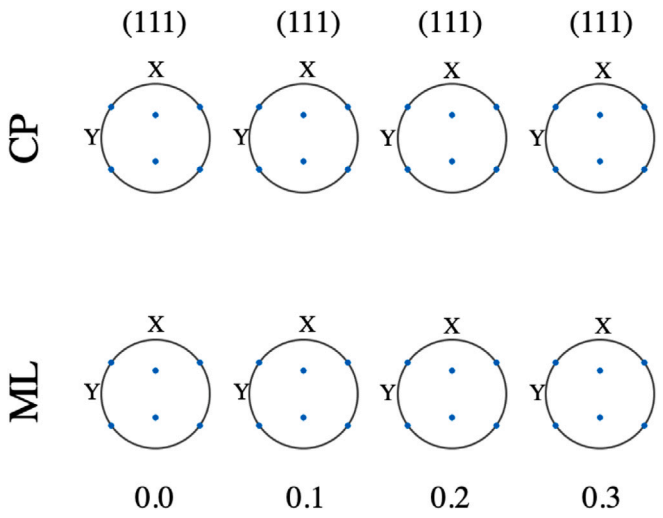


Fig. 10. Comparison between crystal plasticity (CP) and machine learning framework (ML) Goss crystal ⟨111⟩ pole figures displaying texture evolution under compression. The result comparison is indicated for 0.0 (initial texture), 0.1, 0.2, and 0.3 values of $|\varepsilon_{11}|$ engineering strain.

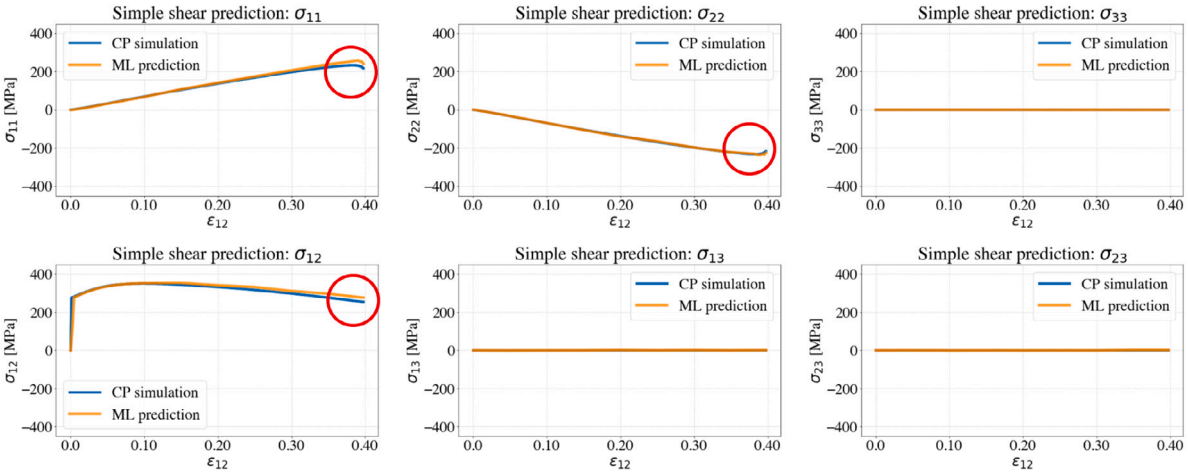


Fig. 11. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) single crystal stress–strain curves under simple shear.

3.2. Predictions for non-monotonic strain paths

To further verify the framework, non-monotonic strain paths were chosen to demonstrate the predictive capabilities of the framework. The non-monotonic strain paths included three different cyclic loadings and an arbitrary non-monotonic loading. The arbitrary loading was also performed on the arbitrary crystal orientation. Note that, the ANN was never trained on data that is non-monotonic and the predictions are obtained using the update of CP parameters as outlined in the manuscript.

Fig. 15 presents the crystal plasticity stress–strain response and the framework predictions for three different types of cyclic loadings. For brevity, only the main stress–strain components are displayed in this work. The results present tension–compression–tension (TCT) cycle simulated for Brass (35°, 45°, 0°) single crystal, compression–tension–compression (CTC) cycle simulated for Taylor (90°, 27°, 45°) single crystal, and cyclic shear was simulated Cube (0°, 0°, 0°). In all the cases, the results are in good agreement with crystal plasticity simulations. The overall prediction error for TCT loading was 5.52 MPa, the prediction error for CTC was 6.90 MPa, and for cyclic shear the error was 3.62 MPa. All the errors are summarised in Table 8. Despite the fact that the errors are higher that the errors for monotonic loadings, the stress–strain behaviour is captured accurately for all cases of cyclic loadings.

The texture evolution predictions were also compared with CP predictions. Figs. 16, 17, and 18 presents the texture evolution for TCT, CTC, and cyclic shear correspondingly. In the cases of TCT and CTC the textures do not exhibit drastic change due to the nature of the strain paths. The predictions were accurate for both cases: the prediction error for TCT was 0.12°, and the prediction error

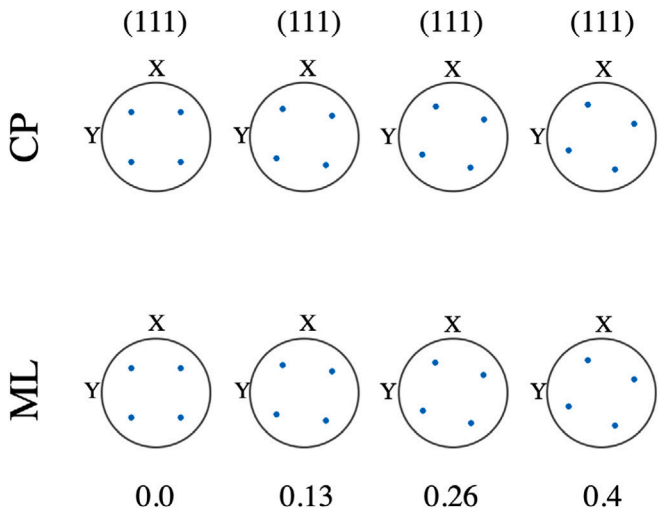


Fig. 12. Comparison between crystal plasticity (CP) and machine learning framework (ML) Cube crystal ⟨111⟩ pole figures displaying texture evolution under simple shear. The result comparison is indicated for 0.0 (initial texture), 0.13, 0.26, and 0.4 values of ε_{12} engineering strain.

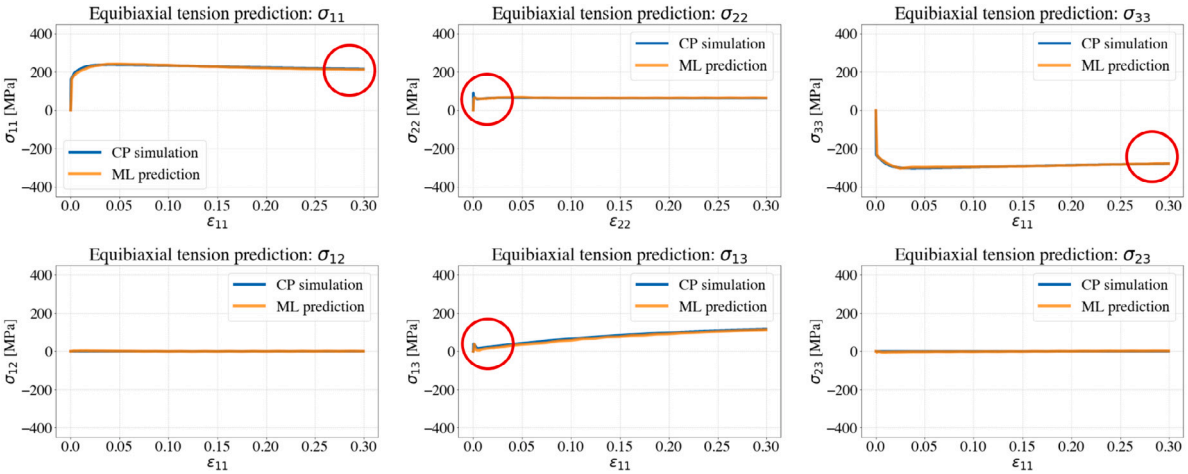


Fig. 13. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) single crystal stress–strain curves under equibiaxial tension.

Table 6
Strain path for the arbitrary loading (single crystal).

Strain-rate tensor, $[D_{11}, D_{22}, D_{33}, D_{12}]$	Number of timesteps ($\Delta t = 0.001$)
[1.0, −0.5, −0.5, −0.3]	50
[0.7, 0.3, −1.0, 0.2]	50
[0.5, −0.3, −0.2, −0.25]	50
[0.7, −0.3, −0.4, −0.25]	50
[0.4, 0.3, −0.7, 0.25]	50
[−0.2, −0.6, 0.8, 0.0]	50
[0.0, 0.0, 0.0, 1.0]	100

for CTC was 0.23°. Fig. 18 displays clearly visible rotation of texture components in the case of cyclic shear. The texture evolution is observed to rotate clockwise during forward shear, and then rotate counter clockwise when the shearing is applied in the other direction. These predictions also agree well with experimental data (Muhammad et al., 2017). To highlight the importance of these results, it should be noted that no training on non-monotonic loading was performed, and the CP-update algorithm enables the framework to predict these non-monotonic strain paths. The error for cyclic shear texture prediction was evaluated at 0.34°, thus, confirming the accuracy of the framework. Error values for texture and stress are also reported in the Table 8.

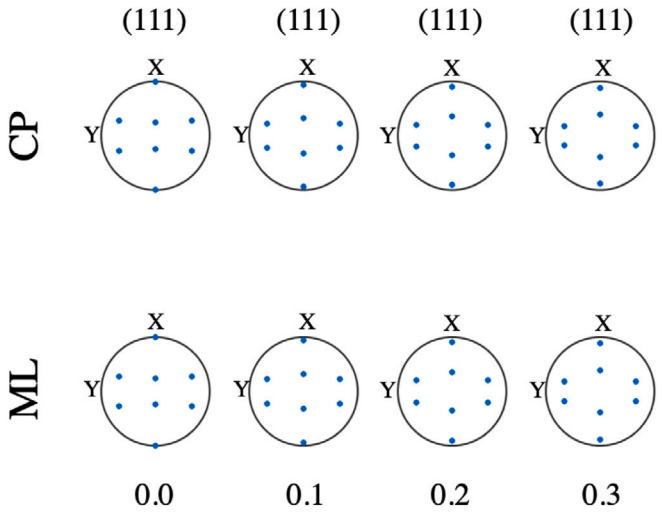


Fig. 14. Comparison between crystal plasticity (CP) and machine learning framework (ML) Copper crystal ⟨111⟩ pole figures displaying texture evolution under equibiaxial tension. The result comparison is indicated for 0.0 (initial texture), 0.1, 0.2, and 0.3 values of ε_{11} engineering strain.

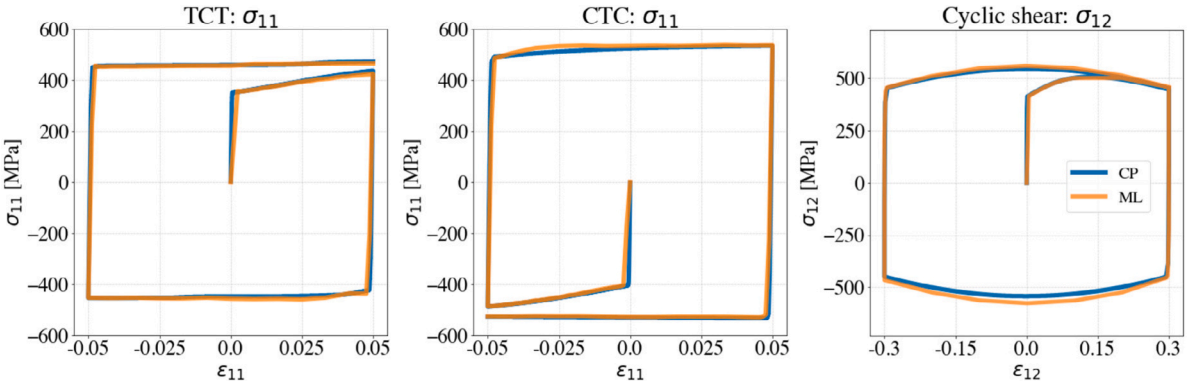


Fig. 15. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) single crystal stress–strain curves under TCT, CTC, and cyclic shear deformation modes.

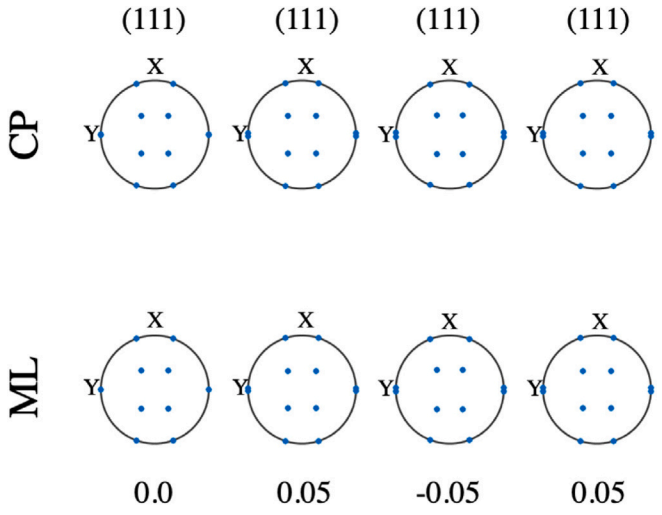


Fig. 16. Comparison between crystal plasticity (CP) and machine learning framework (ML) Brass crystal ⟨111⟩ pole figures displaying texture evolution under TCT deformation mode. The result comparison is indicated for 0.0 (initial texture), 0.05, −0.05, and 0.05 values of ε_{11} engineering strain.

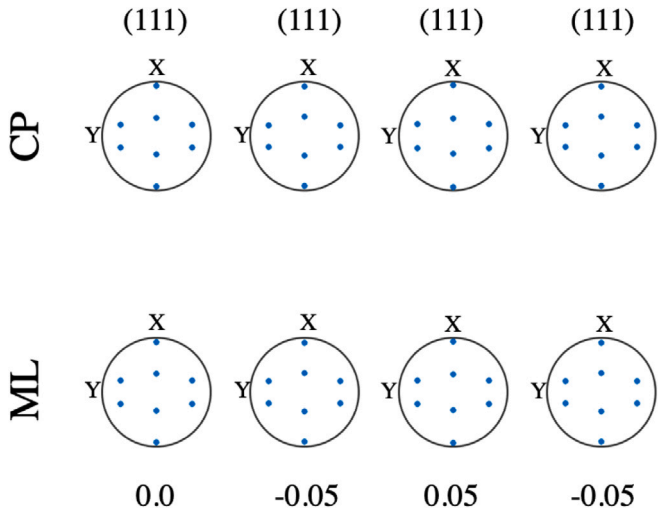


Fig. 17. Comparison between crystal plasticity (CP) and machine learning framework (ML) Taylor crystal ⟨111⟩ pole figures displaying texture evolution under CTC deformation mode. The result comparison is indicated for 0.0 (initial texture), −0.05, 0.05, and −0.05 values of ε_{11} engineering strain.

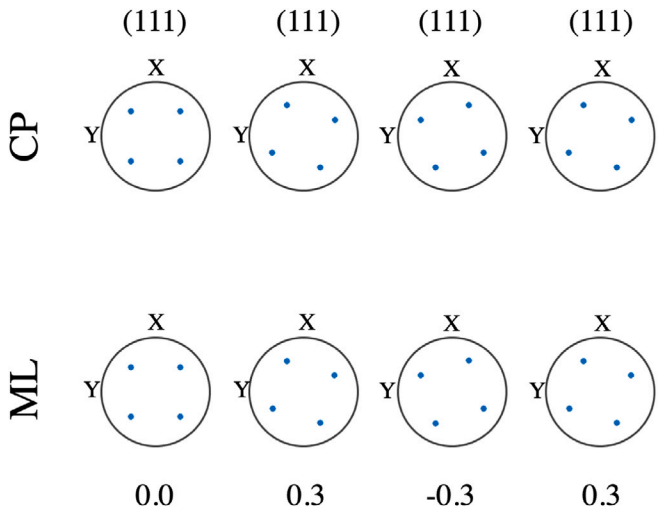


Fig. 18. Comparison between crystal plasticity (CP) and machine learning framework (ML) Cube crystal ⟨111⟩ pole figures displaying texture evolution under cyclic shear deformation mode. The result comparison is indicated for 0.0 (initial texture), 0.3, −0.3, and 0.3 values of ε_{12} engineering strain. A reference axis was included in the pole figures to help better observe texture rotation.

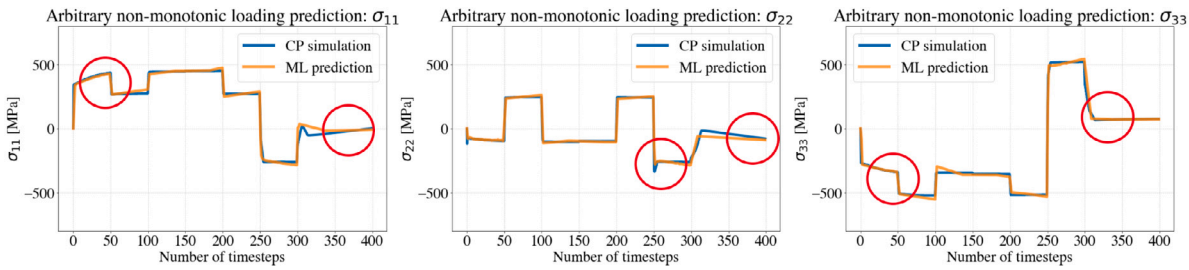


Fig. 19. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) single crystal stress–strain curves under arbitrary non-monotonic loading.

In order to fully verify the predictive capabilities of the framework, a randomly oriented crystal was simulated under a random strain path. The initial texture orientation was taken as (30°, 45°, 45°), and the arbitrary non-monotonic loading consisted of series

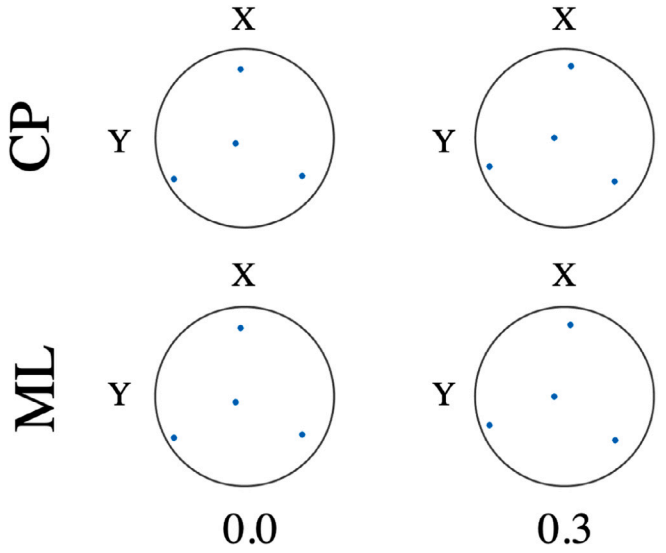


Fig. 20. Comparison between crystal plasticity (CP) and machine learning framework (ML) ⟨111⟩ pole figures for an arbitrary single crystal displaying texture evolution under arbitrary non-monotonic loading for a single crystal with random texture. The result comparison is indicated for initial and final equivalent plastic strain.

of different loading and unloading steps. The applied loading path is summarised in Table 6. The table shows strain path in the form of strain rate tensor and the number of timesteps with a fixed Δt. As can be seen in both Table 7 and flow curve shown in Fig. 19, the single crystal is subjected to 7 different strain paths, chosen randomly. For this strain path, the total amount of equivalent strain during the deformation was equal to 0.3.

The CP simulation was compared with the framework predictions and the comparison is presented in Fig. 19. For brevity, only the principal stress-tensor components are displayed on the plot. The proposed framework was fully able to accurately capture all the subsequent changes in stress values, and results in accurate prediction of final value of stress. The overall prediction for stress values was equal to 3.56 MPa and is reported in Table 8. As a note, the results were obtained with the help of ML models that were trained on only monotonic loadings examples, and no additional training was performed. There is some inaccuracy in prediction, for example for σ_{11} component, but overall the stress–strain behaviour is captured very accurately. Finally, the texture evolution prediction for arbitrary strain path were also well-predicted by the framework. The texture evolution predictions compared with CP simulations can be observed on Fig. 20. Due to the imposed shear strain, some rotation is present in the texture evolution, and it is well-predicted by the framework. The error for texture evolution prediction is 0.62°, thus confirming the accuracy of the framework.

To conclude, in this subsection different non-monotonic strain path predictions were performed for different FCC single crystals. The results were in a good agreement with crystal plasticity simulations and have verified the accuracy of the framework. Therefore, the framework could be applied to predict the polycrystal behaviour under complex strain path. The polycrystal application is discussed in the next subsection.

3.3. Predictions for polycrystals

Results for monotonic and non-monotonic strain paths demonstrated the predictive capabilities of the proposed framework. In this section, the framework was employed for prediction of a polycrystal material response. First, two distinct non-monotonic loadings were compared with CP simulations for a polycrystal constituent of 20% of Cube-oriented crystals, 10% of Goss-oriented crystals, 45% of S-oriented crystal, 20% of Copper-oriented crystals, and 5% of Brass-oriented crystals. Such polycrystal was chosen, as it consisted of common crystal orientations for FCC materials. Then, the framework was used to predict texture evolution of a polycrystal constituent of 246 randomly-oriented grains subjected to plane-strain compression. This case allowed to fully verify the predictive capabilities for texture evolution of the framework. To evaluate the accuracy of the framework, the corresponding CP simulation were performed and used for result comparison. To obtain the polycrystal stress–strain response Taylor type approach (Inal, 2001) was used.

In the first case, the polycrystal was loaded with simple shear up to 30% of shear strain, followed by uniaxial tension up to 30% of uniaxial strain. The deformation was simulated in 600 increments, with $\Delta t = 0.001$. The prediction comparison for all stress tensor components is displayed in Fig. 21. The overall prediction error was evaluated at 6.80 MPa, and is reported in Table 8. The predictions are in a good agreement with crystal plasticity simulations. The σ_{11} component of stress tensor is well-predicted, and the framework captures all the trends correctly. The tension part is predicted exceptionally accurately by the framework. As for the shear stress component, the predictions are in a excellent agreement with crystal plasticity: both shear and tension parts of the loading are predicted correctly. The rest stress components are also well-predicted in their trends and values.

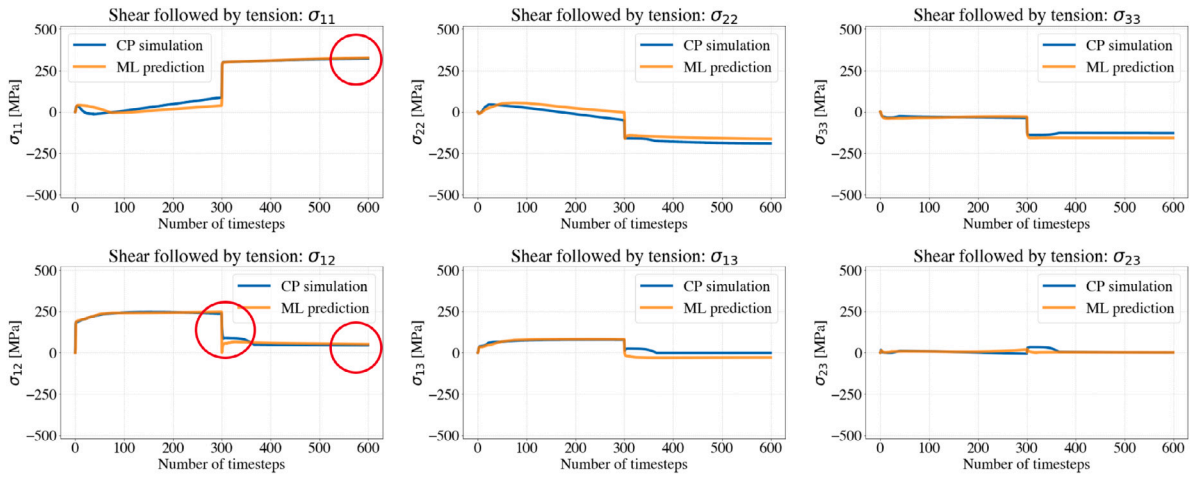


Fig. 21. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) stress–strain predictions for a polycrystal under shear followed by tension strain path.

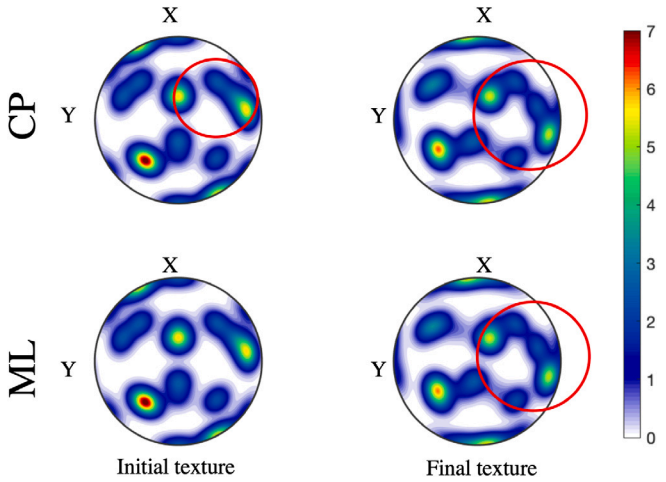


Fig. 22. Comparison between crystal plasticity (CP) and machine learning framework (ML) ⟨111⟩ pole figures displaying texture evolution for a polycrystal under shear followed by tension strain path. The result comparison is indicated for initial and final equivalent plastic strain.

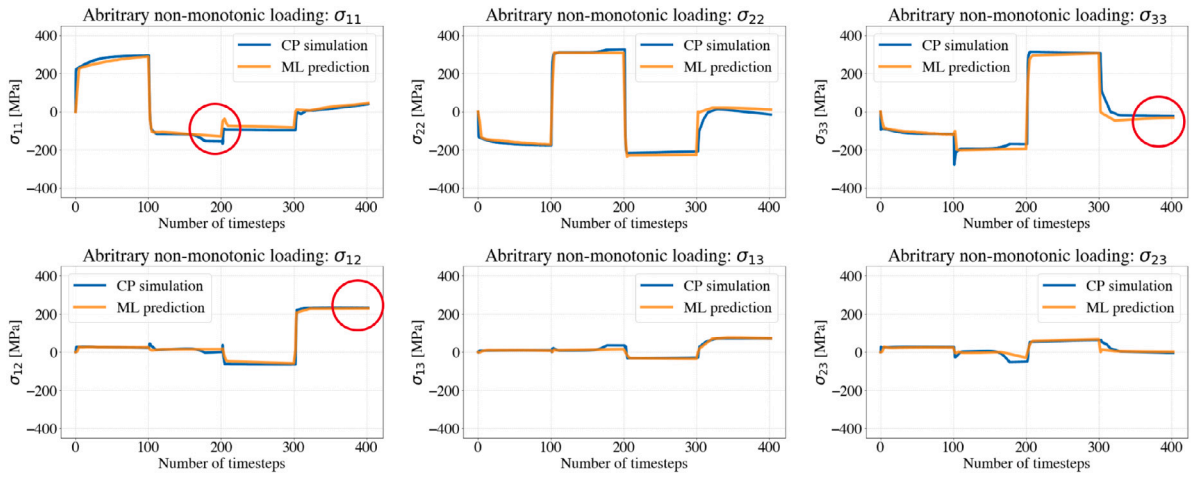


Fig. 23. Comparison between crystal plasticity (CP simulation) and machine learning framework (ML prediction) stress–strain predictions for a polycrystal under arbitrary non-monotonic loading.

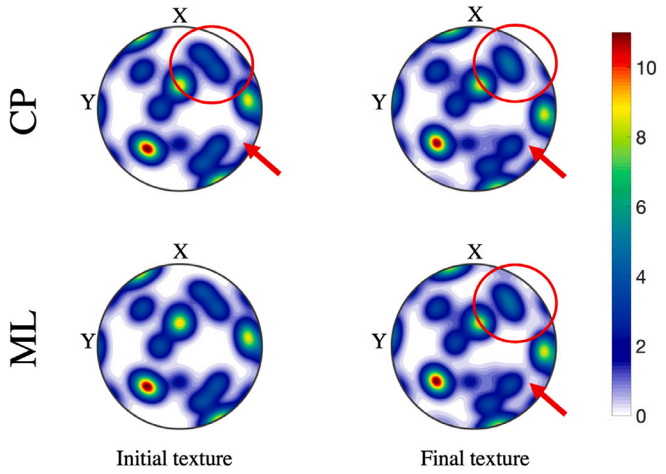


Fig. 24. Comparison between crystal plasticity (CP) and machine learning framework (ML) ⟨111⟩ pole figures displaying texture evolution for a polycrystal under arbitrary non-monotonic loading. The result comparison is indicated for initial and final equivalent plastic strain.

Table 7
Strain path for the arbitrary loading (polycrystal).

Strain-rate tensor, $[D_{11}, D_{22}, D_{33}, D_{12}]$	Number of timesteps ($\Delta t = 0.001$)
[1.0, −0.5, −0.5, 0.0]	100
[−0.3, 1.0, −0.7, 0.2]	100
[−0.1, −0.9, 1.0, 0.25]	100
[0.0, 0.0, 0.0, 1.0]	100

Table 8
Errors for the proposed framework predictions.

Loading mode	Averaged RMSE for stress–strainpredictions (MPa)	Averaged RMSE for texture evolution (°)
Uniaxial tension	1.64	0.0001
Compression	2.02	0.0001
Shear	1.94	0.38
Equibiaxial tension	2.47	0.63
TCT	5.52	0.12
CTC	6.90	0.23
Cyclic shear	3.62	0.34
Arbitrary non-monotonic loading	3.56	0.62
Polycrystal (case 1)	6.80	0.78
Polycrystal (case 2)	5.31	0.75
Polycrystal (case 3)	–	1.44

The texture prediction is shown in Fig. 22. The prediction error for texture is 0.78°, and excellent agreement between the framework prediction and CP simulation is observed. Texture rotation is well-captured by the proposed framework, as well as other changes in texture, as indicated on the plot. The error for texture is also reported in Table 8.

In the second case, the arbitrary non-monotonic loading is applied on the same polycrystal. The exact loading path, that the polycrystal is subjected to, is summarised in Table 7. The polycrystal is subjected to a total of 3 strain path changes with a different strain rate tensor for 100 timesteps, with each timestep having a $\Delta t = 0.001$. Stress–strain comparison is performed for the framework prediction and CP simulation, and is presented in Fig. 23. The overall stress prediction error is equal to 5.31 MPa, and is reported in Table 8. The prediction results are in the good agreement with CP simulation. As indicated on the plot, the framework accurately captured the changes in stresses, resulting in accurate predictions. All the components for stress are captured very accurately, including the sudden changes in stress values for σ_{11}, σ_{12}, σ_{13} components, as indicated on the plot.

Texture prediction for the prediction of arbitrary loading case is also in a good agreement with CP simulation. Fig. 24 presents the pole figures displaying the predictions. The texture prediction error was evaluated at 0.75°, and it is reported in Table 8. Due to imposed shear strain during loading, the rotation is visible on the pole figure and is captured by the framework. As indicated on the plot, the other details are also well-predicted by the framework.

Finally, texture evolution for a random polycrystal subjected to plane-strain compression with 50% reduction in thickness was predicted by the framework and compared to the CP simulation. This strain path was chosen as it is the major strain path during rolling which is a widely investigated process in sheet metal forming. The simulated polycrystal consisted of 246 randomly oriented

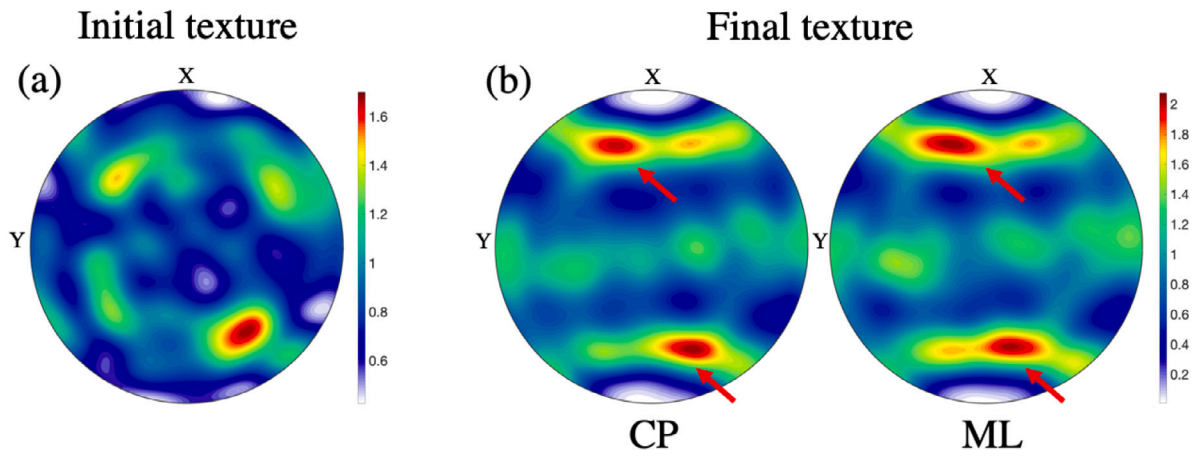


Fig. 25. Comparison between crystal plasticity (CP) and machine learning framework (ML) ⟨111⟩ pole figures showing texture evolution for a polycrystal with a random texture under plane-strain compression, (a) initial texture, (b) and CP and ML comparison for final texture after 50% thickness reduction.

single crystals. Texture prediction using ANN showed an excellent agreement with the crystal plasticity simulation, as demonstrated by the results given in Fig. 25. The framework successfully captures the development of two strong texture components, as pointed at in Fig. 25. The weaker texture band that can be observed in the middle of the final pole figure is also predicted accurately by the framework. The RMSE of the framework prediction in comparison to crystal plasticity simulation was 1.44° and is reported in Table 8.

To conclude, the framework was validated for polycrystal prediction and has shown great predictive capabilities for stress and texture evolution prediction. The prediction were performed for monotonic and non-monotonic strain paths, and were in a good agreement with corresponding CP simulations. The ML models within the framework were trained on only monotonic loadings, and the implemented CP-update algorithm enabled the framework to perform accurate calculations for FCC-family crystals under any non-monotonic strain path.

3.4. Runtime comparison

The results thus far demonstrated the predictive capability of the proposed framework. In addition, compared to crystal plasticity formulations, machine learning models are very efficient computationally. Crystal plasticity simulations, while being accurate in predictions of material behaviour (Brahme et al., 2011b), are computationally expensive due to solving highly-nonlinear differential equations. As mentioned in the introduction, there is a growing body of literature addressing different methods regarding acceleration of crystal plasticity. Therefore, the proposed framework and crystal plasticity runtime comparisons were performed.

Crystal plasticity models could require a substantial number of elements for accurate predictions (Brahme et al., 2012). In addition to that, an exponential increase in computational time with an increase in elements is observed (Ali et al., 2019). Due to this fact, to perform the time comparison, crystal plasticity simulations were performed for 250,000 individual crystal simulations. The crystal plasticity model simulation time was **1** h **and 20** min **on 32 parallel processes**, each running in separate central processing unit core. The ANN framework only required one graphic processing unit, and computation time was **8.5** s. Hence, the proposed framework results in a **computational time savings of over than 99.9**%.

To conclude, the results presented in this section highlight the high computational efficiency of the proposed framework. The framework, consisting of an ensemble of artificial neural networks and crystal plasticity based algorithm, has been shown to produce accurate crystal plasticity predictions of stress–strain properties and texture evolution of the deformed material, and demonstrated tremendous time savings in comparison to crystal plasticity model. One of the most important highlights of the demonstrated results was the ability to predict deformation behaviour on any strain path using the dataset constructed of monotonic-loading examples of deformation.

4. Summary & conclusions

In this paper, a machine learning- and crystal plasticity-based framework is proposed to model stress–strain and texture evolution for a wide variety of the FCC family crystals under non-monotonic strain path. The primary goal of this work is to demonstrate the possibility to obtain accurate and fast crystal plasticity predictions for non-monotonic strain path with the aid of machine learning, while not needing to train the machine learning models on the infinite set of possible loading conditions. The approach of coupling crystal plasticity and machine learning is feasible for training, accurate, and is computationally efficient in its application. As far as the authors are aware, this is the first study that demonstrates the machine learning-based framework capable of predicting the full flow stress behaviour and texture evolution of FCC crystals under any strain path for any initial texture configuration. The important observations and conclusions are as follows:

- Typically, machine learning methods are only useful in building the mappings that generalise predictions within the bounding box of the available dataset. These methods usually perform poorly on extrapolation tasks. The proposed framework consists of an ensemble of artificial neural networks trained only on monotonic loading examples. The implemented crystal plasticity-based algorithm makes it possible to predict the complex cases of non-monotonic loadings using those networks. The presented framework allows accurate material behaviour predictions on any, monotonic or non-monotonic, strain path. The predictions include full stress–strain predictions and texture evolution.
- The present application of ML for crystal plasticity involves training features of varied complexity. For instance, certain training features, such as the rotation matrix, evolve in a linear manner and other features, such as the stress tensor, can exhibit highly non-linear behaviour. This complexity and non-linearity can be modelled accurately using FFNN's with the proper choice of the number of layers, unit cells within each layer, and activation functions. Furthermore, the numerical implementation of FFNNs is relatively simpler compared to other complex methods such as recurrent neural networks.
- The proposed machine learning- and crystal plasticity-based framework has been demonstrated to accurately predict the stress–strain and texture evolution for different single crystals for the a wide variety of strain paths: uniaxial tension, compression, simple shear, equibiaxial tension, tension–compression–tension, compression–tension–compression, cyclic shear, and arbitrary non-monotonic loading.
- The framework demonstrated excellent predictive capabilities for polycrystal simulations. The two cases were considered in this research: shear followed by tension, and arbitrary non-monotonic loading. For the first case, the stress–strain prediction error was 6.80 MPa and the texture evolution error was 0.78°. For the second case, the stress–strain prediction error was 5.31 MPa and the texture evolution error was 0.75°. In both cases, the prediction error is acceptable and validates the accuracy of the framework.
- The framework demonstrated excellent ability to predict texture evolution under plane-strain compression which is the major strain path during rolling of sheet metals. The overall RMSE error between CP and ANN texture predictions was 1.44°, confirming the capability and accuracy of the proposed framework to predict texture evolution during large strain plasticity applications.
- In other cases for monotonic and non-monotonic loadings predictions, the overall RMSE for stress–strain predictions did not exceed 10 MPa, and the overall RMSE for texture evolution did not exceed 1°. This error is of within acceptable values and, thus, verifies the accuracy of the proposed framework.
- In addition to accuracy, the framework is capable of prediction of subtle changes of values in stress behaviour. Texture predictions by the framework were in excellent agreement with CP simulations for most common orientations of FCC family crystals, and for random textures as well.
- The presented approach is computationally efficient and shows up to 99.9% computational speedup over conventional crystal plasticity model. As a direction to further research, the framework can be used to make predictions for a coupon- or component-level deformation, which was not feasible in the past due to high computational demands of the crystal plasticity model.
- One of the most important outcomes of the current work is the proof of feasibility: it is possible to utilise machine learning to capture the crystal plasticity predictions of stress–strain properties and texture evolution. The presented results suggest that machine learning methods can be successfully applied to perform crystal plasticity prediction of material deformation behaviour.

CRediT authorship contribution statement

Olga Ibragimova: Conceptualisation, Investigation, Methodology, Formal analysis, Software, Visualisation, Writing - original draft. **Abhijit Brahme:** Conceptualisation, Investigation, Methodology, Supervision, Writing - review & editing. **Waqas Muhammad:** Conceptualisation, Investigation, Methodology, Supervision, Writing - review & editing. **Julie Lévesque:** Supervision, Project administration, Funding acquisition. **Kaan Inal:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge Quebec Ministry of Economy and Innovation, Canada, via the CQRDA (project number 1066), Natural Sciences and Engineering Research Council of Canada (project number 531057-2018), and Verbom, Canada for supporting this work. The authors are thankful to Dr. Christopher Kohar at the University of Waterloo for his technical support with the high-performance computing cluster that was used to support this work.

References

Abedini, A., Butcher, C., Worswick, M.J., 2018. Experimental fracture characterisation of an anisotropic magnesium alloy sheet in proportional and non-proportional loading conditions. Int. J. Solids Struct. 144, 1–19.

Abiodun, O.I., Jantan, A., Omolara, A.E., Dada, K.V., Mohamed, N.A., Arshad, H., 2018. State-of-the-art in artificial neural network applications: A survey. Heliyon 4 (11), e00938.

Acar, P., 2020. Machine learning reinforced crystal plasticity modeling under experimental uncertainty. In: AIAA Scitech 2020 Forum. p. 1152.

Alankar, A., Eisenlohr, P., Raabe, D., 2011. A dislocation density-based crystal plasticity constitutive model for prismatic slip in α-titanium. Acta Mater. 59 (18), 7003–7009.

Ali, U., Muhammad, W., Brahme, A., Skiba, O., Inal, K., 2019. Application of artificial neural networks in micromechanics for polycrystalline metals. Int. J. Plast. 120, 205–219.

Ali, U., Odoh, D., Muhammad, W., Brahme, A., Mishra, R.K., Wells, M., Inal, K., 2017. Experimental investigation and through process crystal plasticity-static recrystallization modeling of temperature and strain rate effects during hot compression of AA6063. Mater. Sci. Eng. A 700, 374–386.

Arsenlis, A., Parks, D.M., 2002. Modeling the evolution of crystallographic dislocation density in crystal plasticity. J. Mech. Phys. Solids 50 (9), 1979–2009.

Asaro, R.J., Needleman, A., 1985. Overview no. 42 texture development and strain hardening in rate dependent polycrystals. Acta Metall. 33 (6), 923–953.

Banovic, S., Iadicola, M., Foecke, T., 2008. Textural development of AA 5754 sheet deformed under in-plane biaxial tension. Metall. Mater. Trans. A 39 (9), 2246.

Bessa, M., Bostanabad, R., Liu, Z., Hu, A., Apley, D.W., Brinson, C., Chen, W., Liu, W.K., 2017. A framework for data-driven analysis of materials under uncertainty: Countering the curse of dimensionality. Comput. Methods Appl. Mech. Engrg. 320, 633–667.

Bishop, C.M., 1994. Neural networks and their applications. Rev. Sci. Instrum. 65 (6), 1803–1832.

Bishop, C.M., 2006. Pattern Recognition and Machine Learning. springer.

Brahme, A.P., Inal, K., Mishra, R.K., Saimoto, S., 2011a. The backstress effect of evolving deformation boundaries in fcc polycrystals. Int. J. Plast. 27 (8), 1252–1266.

Brahme, A.P., Inal, K., Mishra, R.K., Saimoto, S., 2011b. A new strain hardening model for rate-dependent crystal plasticity. Comput. Mater. Sci. 50 (10), 2898–2908.

Brahme, A., Staraselski, Y., Inal, K., Mishra, R.K., 2012. Determination of the minimum scan size to obtain representative textures by electron backscatter diffraction. Metall. Mater. Trans. A 43 (13), 5298–5307.

Brahme, A., Winning, M., Raabe, D., 2009. Prediction of cold rolling texture of steels using an artificial neural network. Comput. Mater. Sci. 46 (4), 800–804.

Bratley, P., Fox, B.L., 1988. Algorithm 659: Implementing Sobol's quasirandom sequence generator. ACM Trans. Math. Softw. (TOMS) 14 (1), 88–100.

Brough, D.B., Wheeler, D., Kalidindi, S.R., 2017. Materials knowledge systems in python—a data science framework for accelerated development of hierarchical materials. Integr. Mater. Manuf. Innov. 6 (1), 36–53.

Burgess, J., Gallagher, B., Jensen, D.D., Levine, B.N., et al., 2006. MaxProp: Routing for vehicle-based disruption-tolerant networks. In: Infocom, Vol. 6. Barcelona, Spain.

Burhenne, S., Jacob, D., Henze, G.P., 2011. Sampling based on Sobol'sequences for Monte Carlo techniques applied to building simulations. In: Proc. Int. Conf. Build. Simulat. pp. 1816–1823.

Chakraborty, P., Joseph, D.S., Ghosh, S., 2011. Wavelet transformation based multi-time scale crystal plasticity FEM for cyclic deformation in titanium alloys under dwell load. Finite Elem. Anal. Des. 47 (6), 610–618.

Chang, Y.W., Asaro, R.J., 1981. An experimental study of shear localization in aluminum-copper single crystals. Acta Metall. 29 (1), 241–257.

Cho, K., Van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y., 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation. arXiv preprint arXiv:1406.1078.

Chollet, F., 2018. Deep Learning Mit Python Und Keras: Das Praxis-Handbuch Vom Entwickler Der Keras-Bibliothek. MITP-Verlags GmbH & Co. KG.

Cormack, G.V., Hidalgo, J.M.G., Sánz, E.P., 2007. Feature engineering for mobile (SMS) spam filtering. In: Proceedings of the 30th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval. ACM, pp. 871–872.

Cyr, E., Mohammadi, M., Brahme, A., Mishra, R.K., Inal, K., 2017. Modeling the formability of aluminum alloys at elevated temperatures using a new thermo-elasto-viscoplastic crystal plasticity framework. Int. J. Mech. Sci. 128, 312–325.

Eghtesad, A., Zecevic, M., Lebensohn, R.A., McCabe, R.J., Knezevic, M., 2018. Spectral database constitutive representation within a spectral micromechanical solver for computationally efficient polycrystal plasticity modelling. Comput. Mech. 61 (1–2), 89–104.

Evers, L., Brekelmans, W., Geers, M., 2004. Scale dependent crystal plasticity framework with dislocation density and grain boundary effects. Int. J. Solids Struct. 41 (18–19), 5209–5230.

Fan, X., Yang, H., 2011. Internal-state-variable based self-consistent constitutive modeling for hot working of two-phase titanium alloys coupling microstructure evolution. Int. J. Plast. 27 (11), 1833–1852.

Farooq, H., Cailletaud, G., Forest, S., Ryckelynck, D., 2020. Crystal plasticity modeling of the cyclic behavior of polycrystalline aggregates under non-symmetric uniaxial loading: Global and local analyses. Int. J. Plast. 126, 102619.

Feather, W.G., Ghorbanpour, S., Savage, D.J., Ardeljan, M., Jahedi, M., McWilliams, B.A., Gupta, N., Xiang, C., Vogel, S.C., Knezevic, M., 2019. Mechanical response, twinning, and texture evolution of WE43 magnesium-rare earth alloy as a function of strain rate: experiments and multi-level crystal plasticity modeling. Int. J. Plast. 120, 180–204.

Folk, M., Heber, G., Koziol, Q., Pourmal, E., Robinson, D., 2011. An overview of the HDF5 technology suite and its applications. In: Proceedings of the EDBT/ICDT 2011 Workshop on Array Databases. ACM, pp. 36–47.

Follansbee, P., Kocks, U., 1988. A constitutive description of the deformation of copper based on the use of the mechanical threshold stress as an internal state variable. Acta Metall. 36 (1), 81–93.

Friedman, J., Hastie, T., Tibshirani, R., 2001. The Elements of Statistical Learning, Vol. 1. Springer series in statistics New York.

Garla, V.N., Brandt, C., 2012. Ontology-guided feature engineering for clinical text classification. J. Biomed. Inform. 45 (5), 992–998.

Ghaboussi, J., Garrett Jr, J., Wu, X., 1991. Knowledge-based modeling of material behavior with neural networks. J. Eng. Mech. 117 (1), 132–153.

Ghaboussi, J., Sidarta, D., 1998. New nested adaptive neural networks (NANN) for constitutive modeling. Comput. Geotech. 22 (1), 29–52.

Gorji, M.B., Mozaffar, M., Heidenreich, J.N., Cao, J., Mohr, D., 2020. On the potential of recurrent neural networks for modeling path dependent plasticity. J. Mech. Phys. Solids 103972.

Graves, A., Jaitly, N., Mohamed, A.-r., 2013. Hybrid speech recognition with deep bidirectional LSTM. In: 2013 IEEE Workshop on Automatic Speech Recognition and Understanding. IEEE, pp. 273–278.

Gupta, A., Bettaieb, M.B., Abed-Meraim, F., Kalidindi, S.R., 2018. Computationally efficient predictions of crystal plasticity based forming limit diagrams using a spectral database. Int. J. Plast. 103, 168–187.

Haghdadi, N., Zarei-Hanzaki, A., Khalesian, A., Abedi, H., 2013. Artificial neural network modeling to predict the hot deformation behavior of an A356 aluminum alloy. Mater. Des. 49, 386–391.

Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Comput. 9 (8), 1735–1780.

Hornik, K., Stinchcombe, M., White, H., 1989. Multilayer feedforward networks are universal approximators. Neural Netw. 2 (5), 359–366.

Horstemeyer, M.F., Bammann, D.J., 2010. Historical review of internal state variable theory for inelasticity. Int. J. Plast. 26 (9), 1310–1334.
Inal, K., 2001. Numerical Simulation of Sheet Metal Forming Processes and Localized Deformation Phenomena for FCC Polycrystals (Ph.D. thesis). Université de Sherbrooke, Sherbrooke, QC.
Inal, K., Neale, K., Aboutajeddine, A., 2005. Forming limit comparisons for FCC and BCC sheets. Int. J. Plast. 21 (6), 1255–1266.
Inal, K., Wu, P., Neale, K., 2002. Finite element analysis of localization in FCC polycrystalline sheets under plane stress tension. Int. J. Solids Struct. 39 (13–14), 3469–3486.
Izadbakhsh, A., Inal, K., Mishra, R.K., Niewczas, M., 2011. New crystal plasticity constitutive model for large strain deformation in single crystals of magnesium. Comput. Mater. Sci. 50 (7), 2185–2202.
Jain, A.K., Chandrasekaran, B., 1982. 39 dimensionality and sample size considerations in pattern recognition practice. Handbook of Statist. 2, 835–855.
Joseph, D.S., Chakraborty, P., Ghosh, S., 2010. Wavelet transformation based multi-time scaling method for crystal plasticity FE simulations under cyclic loading. Comput. Methods Appl. Mech. Engrg. 199 (33–36), 2177–2194.
Kalidindi, S.R., 2012. Computationally efficient, fully coupled multiscale modeling of materials phenomena using calibrated localization linkages. ISRN Mater. Sci. 2012.
Kalidindi, S.R., 2013. Microstructure informatics. In: Informatics for Materials Science and Engineering. Elsevier, pp. 443–466.
Kalidindi, S.R., 2015a. Hierarchical Materials Informatics: Novel Analytics for Materials Data. Elsevier.
Kalidindi, S.R., 2015b. Data science and cyberinfrastructure: critical enablers for accelerated development of hierarchical materials. Int. Mater. Rev. 60 (3), 150–168.
Kalidindi, S.R., Gupta, A., Popova, E., 2020. Computationally efficient crystal plasticity simulations using spectral databases. In: Handbook of Materials Modeling: Methods: Theory and Modeling. Springer, pp. 1685–1710.
Kalidindi, S.R., Niezgoda, S.R., Landi, G., Vachhani, S., Fast, T., 2010. A novel framework for building materials knowledge systems. Comput. Mater. Contin. 17 (2), 103–125.
Keren, G., Schuller, B., 2016. Convolutional RNN: an enhanced model for extracting features from sequential data. In: 2016 International Joint Conference on Neural Networks (IJCNN). IEEE, pp. 3412–3419.
Kingma, D.P., Ba, J., 2014. Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980.
Knezevic, M., Kalidindi, S.R., 2017. Crystal plasticity modeling of microstructure evolution and mechanical fields during processing of metals using spectral databases. JOM 69 (5), 830–838.
Kohar, C.P., Bassani, J.L., Brahme, A., Muhammad, W., Mishra, R.K., Inal, K., 2019. A new multi-scale framework to incorporate microstructure evolution in phenomenological plasticity: theory, explicit finite element formulation, implementation and validation. Int. J. Plast. 117, 122–156.
Koutnik, J., Greff, K., Gomez, F., Schmidhuber, J., 2014. A clockwork RNN. In: International Conference on Machine Learning. PMLR, pp. 1863–1871.
Latypov, M.I., Toth, L.S., Kalidindi, S.R., 2019. Materials knowledge system for nonlinear composites. Comput. Methods Appl. Mech. Engrg. 346, 180–196. http://dx.doi.org/10.1016/j.cma.2018.11.034.
Lefik, M., Schrefler, B.A., 2003. Artificial neural network as an incremental non-linear constitutive model for a finite element code. Comput. Methods Appl. Mech. Engrg. 192 (28–30), 3265–3283.
Li, Y.L., Kohar, C.P., Mishra, R.K., Inal, K., 2020. A new crystal plasticity constitutive model for simulating precipitation-hardenable aluminum alloys. Int. J. Plast. 102759.
Li, X., Roth, C.C., Mohr, D., 2019. Machine-learning based temperature-and rate-dependent plasticity model: application to analysis of fracture experiments on DP steel. Int. J. Plast. 118, 320–344.
Mangal, A., Holm, E.A., 2018a. A dataset of synthetic face centered cubic 3D polycrystalline microstructures, grain-wise microstructural descriptors and grain averaged stress fields under uniaxial tensile deformation. Data Brief 19, 2029–2036. http://dx.doi.org/10.1016/j.dib.2018.06.072.
Mangal, A., Holm, E.A., 2018b. Applied machine learning to predict stress hotspots I: Face centered cubic materials. Int. J. Plast. 111, 122–134.
Mangal, A., Holm, E.A., 2019. Applied machine learning to predict stress hotspots II: Hexagonal close packed materials. Int. J. Plast. 114, 1–14.
Miyazawa, Y., Briffod, F., Shiraiwa, T., Enoki, M., 2019. Prediction of cyclic stress–strain property of steels by crystal plasticity simulations and machine learning. Materials 12 (22), 3668.
Mozaffar, M., Bostanabad, R., Chen, W., Ehmann, K., Cao, J., Bessa, M., 2019. Deep learning predicts path-dependent plasticity. Proc. Natl. Acad. Sci. 116 (52), 26414–26420.
Muhammad, W., Brahme, A.P., Ibragimova, O., Kang, J., Inal, K., 2020. A machine learning framework to predict local strain distribution and the evolution of plastic anisotropy & fracture in additively manufactured alloys. Int. J. Plast. 136, 102867.
Muhammad, W., Brahme, A.P., Kang, J., Mishra, R.K., Inal, K., 2017. Experimental and numerical investigation of texture evolution and the effects of intragranular backstresses in aluminum alloys subjected to large strain cyclic deformation. Int. J. Plast. 93, 137–163.
Nesterov, Y.E., 1983. A method for solving the convex programming problem with convergence rate O (1/kˆ 2). In: Dokl. Akad. Nauk Sssr, Vol. 269. pp. 543–547.
de Oca Zapiain, D.M., Popova, E., Kalidindi, S.R., 2017. Prediction of microscale plastic strain rate fields in two-phase composites subjected to an arbitrary macroscale strain rate using the materials knowledge system framework. Acta Mater. 141, 230–240.
Pandey, A., Khan, A.S., Kim, E.-Y., Choi, S.-H., Gnäupel-Herold, T., 2013. Experimental and numerical investigations of yield surface, texture, and deformation mechanisms in AA5754 over low to high temperatures and strain rates. Int. J. Plast. 41, 165–188.
Pandey, A., Pokharel, R., 2020. Machine learning enabled surrogate crystal plasticity model for spatially resolved 3D orientation evolution under uniaxial tension. arXiv preprint arXiv:2005.00951.
Papanikolaou, S., Tzimas, M., Reid, A.C., Langer, S.A., 2019. Spatial strain correlations, machine learning, and deformation history in crystal plasticity. Phys. Rev. E 99 (5), 053003.
Paramatmuni, C., Kanjarla, A.K., 2019. A crystal plasticity FFT based study of deformation twinning, anisotropy and micromechanics in HCP materials: Application to AZ31 alloy. Int. J. Plast. 113, 269–290.
Perzyna, P., 1986. Internal state variable description of dynamic fracture of ductile solids. Int. J. Solids Struct. 22 (7), 797–818.
Plunkett, B., Cazacu, O., Barlat, F., 2008. Orthotropic yield criteria for description of the anisotropy in tension and compression of sheet metals. Int. J. Plast. 24 (5), 847–866.
Popova, E., Staraselski, Y., Brahme, A., Mishra, R., Inal, K., 2015. Coupled crystal plasticity–probabilistic cellular automata approach to model dynamic recrystallization in magnesium alloys. Int. J. Plast. 66, 85–102.
Raudys, S.J., Jain, A.K., et al., 1991. Small sample size effects in statistical pattern recognition: Recommendations for practitioners. IEEE Trans. Pattern Anal. Mach. Intell. 13 (3), 252–264.
Rossiter, J., Brahme, A., Simha, M., Inal, K., Mishra, R., 2010. A new crystal plasticity scheme for explicit time integration codes to simulate deformation in 3D microstructures: effects of strain path, strain rate and thermal softening on localized deformation in the aluminum alloy 5754 during simple shear. Int. J. Plast. 26 (12), 1702–1725.
Roters, F., Eisenlohr, P., Hantcherli, L., Tjahjanto, D.D., Bieler, T.R., Raabe, D., 2010. Overview of constitutive laws, kinematics, homogenization and multiscale methods in crystal plasticity finite-element modeling: Theory, experiments, applications. Acta Mater. 58 (4), 1152–1211.
Roters, F., Raabe, D., Gottstein, G., 2000. Work hardening in heterogeneous alloys—a microstructural approach based on three internal state variables. Acta Mater. 48 (17), 4181–4189.

Sabokpa, O., Zarei-Hanzaki, A., Abedi, H., Haghdadi, N., 2012. Artificial neural network modeling to predict the high temperature flow behavior of an AZ81 magnesium alloy. Mater. Des. 39, 390–396.
Santiago, J., Claeys-Bruno, M., Sergent, M., 2012. Construction of space-filling designs using WSP algorithm for high dimensional spaces. Chemometr. Intell. Lab. Syst. 113, 26–31.
Seide, F., Li, G., Chen, X., Yu, D., 2011. Feature engineering in context-dependent deep neural networks for conversational speech transcription. In: 2011 IEEE Workshop on Automatic Speech Recognition & Understanding. IEEE, pp. 24–29.
Severyn, A., Moschitti, A., 2013. Automatic feature engineering for answer selection and extraction. In: Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing. pp. 458–467.
Sharma, S., 2017. Activation functions in neural networks. Towards Data Sci. 6.
Sobol', I.M., 1967. On the distribution of points in a cube and the approximate evaluation of integrals. Zh. Vychisl. Mat. Mat. Fiz. 7 (4), 784–802.
Sobol, I.M., 1976. Uniformly distributed sequences with an additional uniform property. USSR Comput. Math. Math. Phys. 16 (5), 236–242.
Tallman, A.E., Swiler, L.P., Wang, Y., McDowell, D.L., 2017. Reconciled top-down and bottom-up hierarchical multiscale calibration of bcc fe crystal plasticity. Int. J. Multiscale Comput. Eng. 15 (6).
Wang, H., Wu, P., Wang, J., Tomé, C., 2013. A crystal plasticity model for hexagonal close packed (HCP) crystals including twinning and de-twinning mechanisms. Int. J. Plast. 49, 36–52.
Xiao, X., Chen, L., Yu, L., Duan, H., 2019. Modelling nano-indentation of ion-irradiated FCC single crystals by strain-gradient crystal plasticity theory. Int. J. Plast. 116, 216–231.
Zadeh, M.R., Amin, S., Khalili, D., Singh, V.P., 2010. Daily outflow prediction by multi layer perceptron with logistic sigmoid and tangent sigmoid activation functions. Water Resour. Manage. 24 (11), 2673–2688.
Zecevic, M., Korkolis, Y.P., Kuwabara, T., Knezevic, M., 2016. Dual-phase steel sheets under cyclic tension–compression to large strains: experiments and crystal plasticity modeling. J. Mech. Phys. Solids 96, 65–87.
Zeiler, M.D., 2012. Adadelta: an adaptive learning rate method. arXiv preprint arXiv:1212.5701.
Zhang, P., Kohar, C.P., Brahme, A.P., Choi, S.-H., Mishra, R.K., Inal, K., 2019. A crystal plasticity formulation for simulating the formability of a transformation induced plasticity steel. J. Mater Process. Technol. 116493.
Zhang, Y., Ling, C., 2018. A strategy to apply machine learning to small datasets in materials science. Npj Comput. Mater. 4 (1), 1–8.